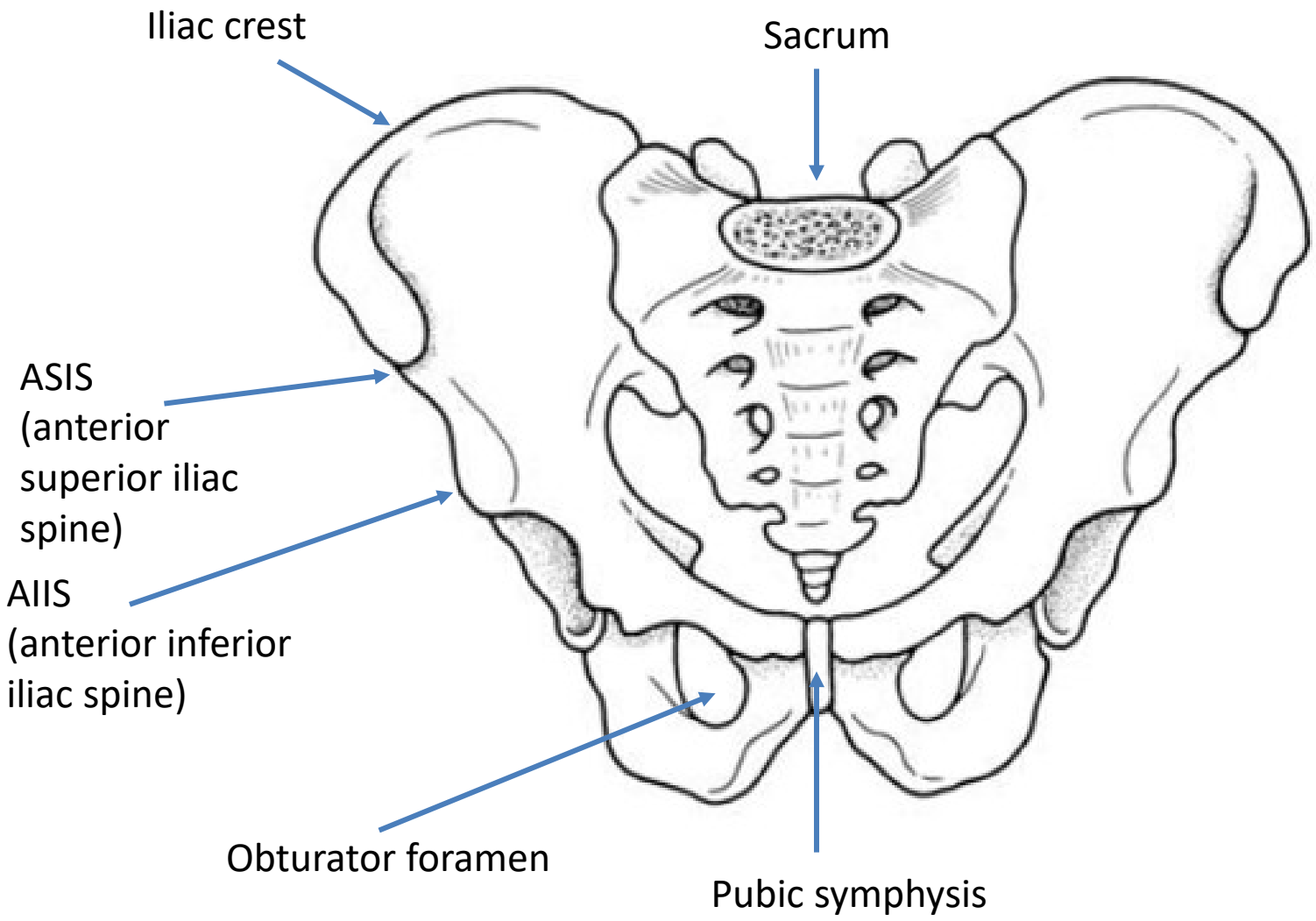
Osteology of Os Coxae and femur

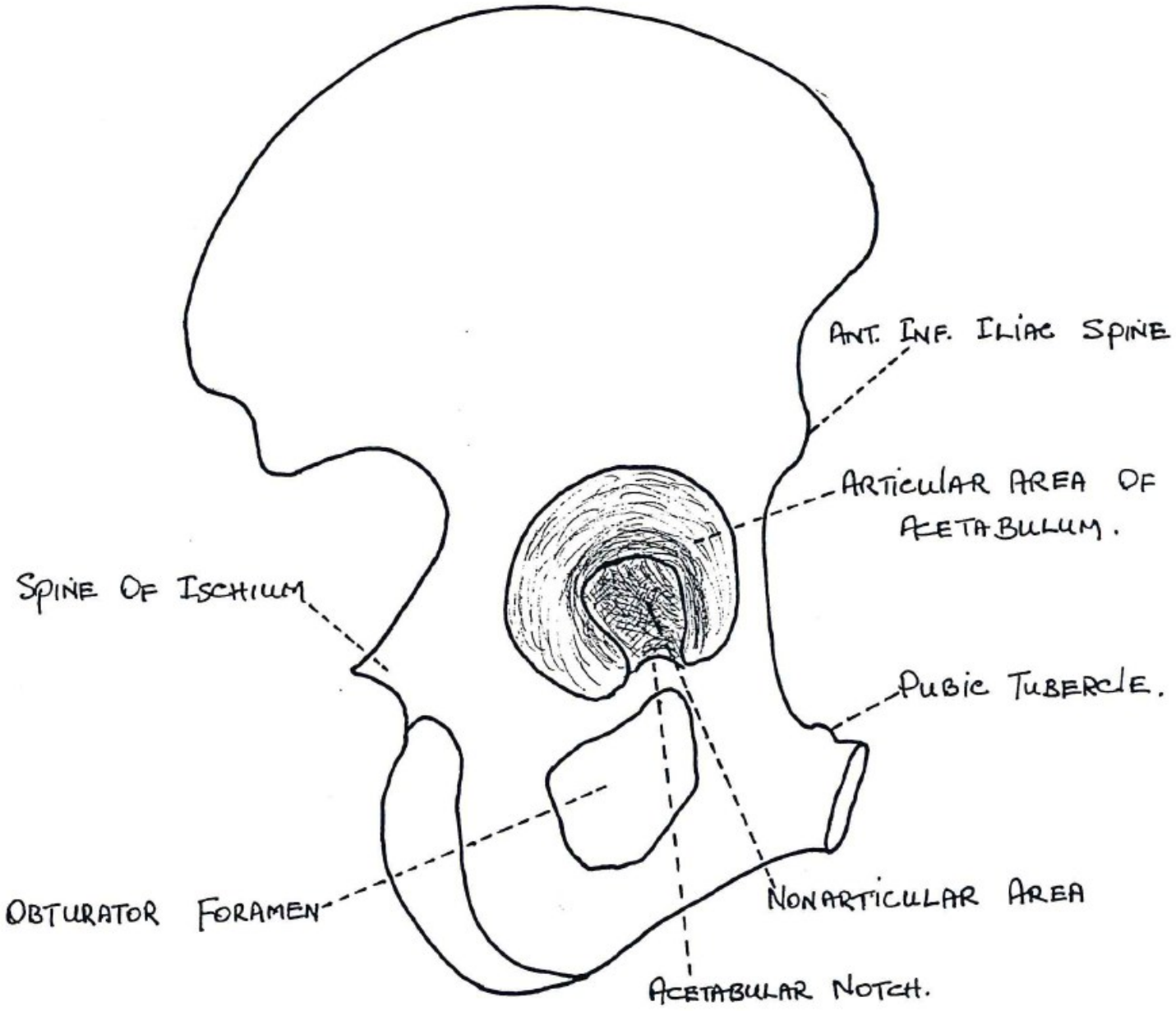
Learning Focus of this Lecture

- Osteology of Pelvis
- Osteology of Femur
- Development of Pelvis including some anomalies
- Sexual dimorphism of pelvis
- Adaptation of female pelvis to parturition
- Radiology of the Hip
- The blood supply to the head of the femur
- Femoral and Q angles
- Clinical examination of hip: Nelaton’s line/Bryant’s triangle

Os Coxae (plural) and sacrum



Hip Bone = Os coxa = Os innominatum



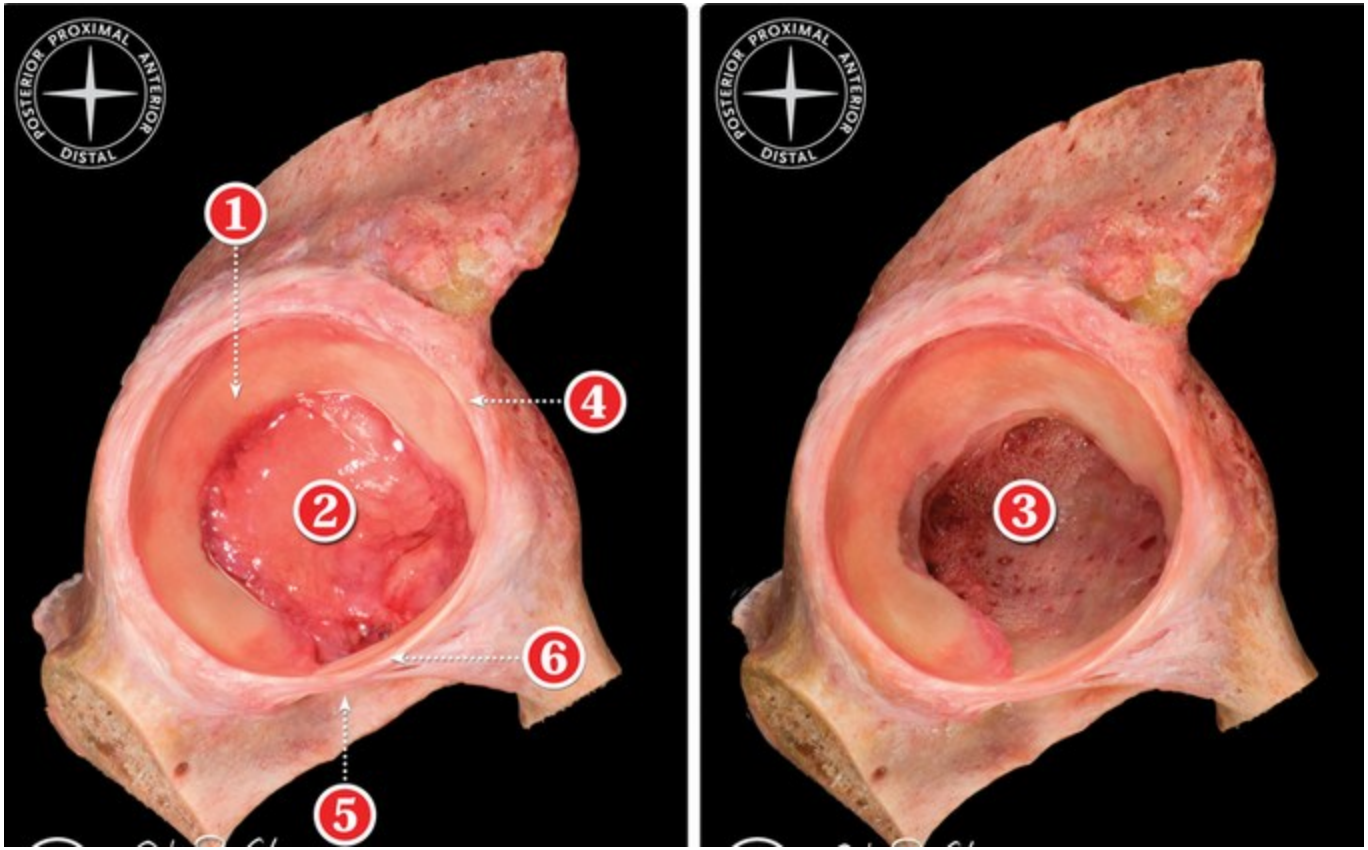
ANATOMY OF THE HIP JOINT

AND HOW IT IS RELATED TO GROIN PAIN

– Written by Miquel Dalmau-Pastor, Jordi Vega and Pau Golanó, Spain

Acetabulum=vinegar cup



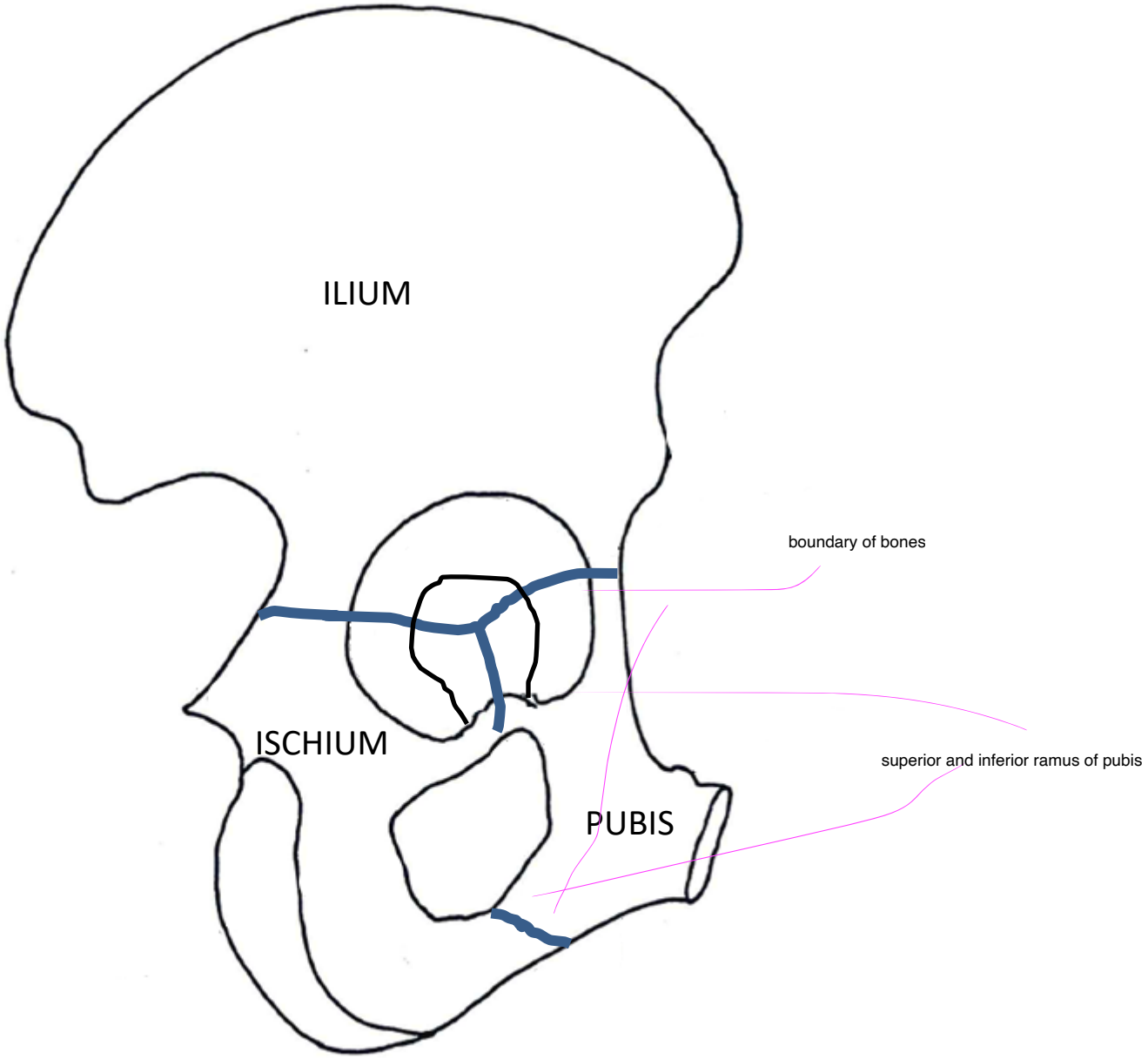


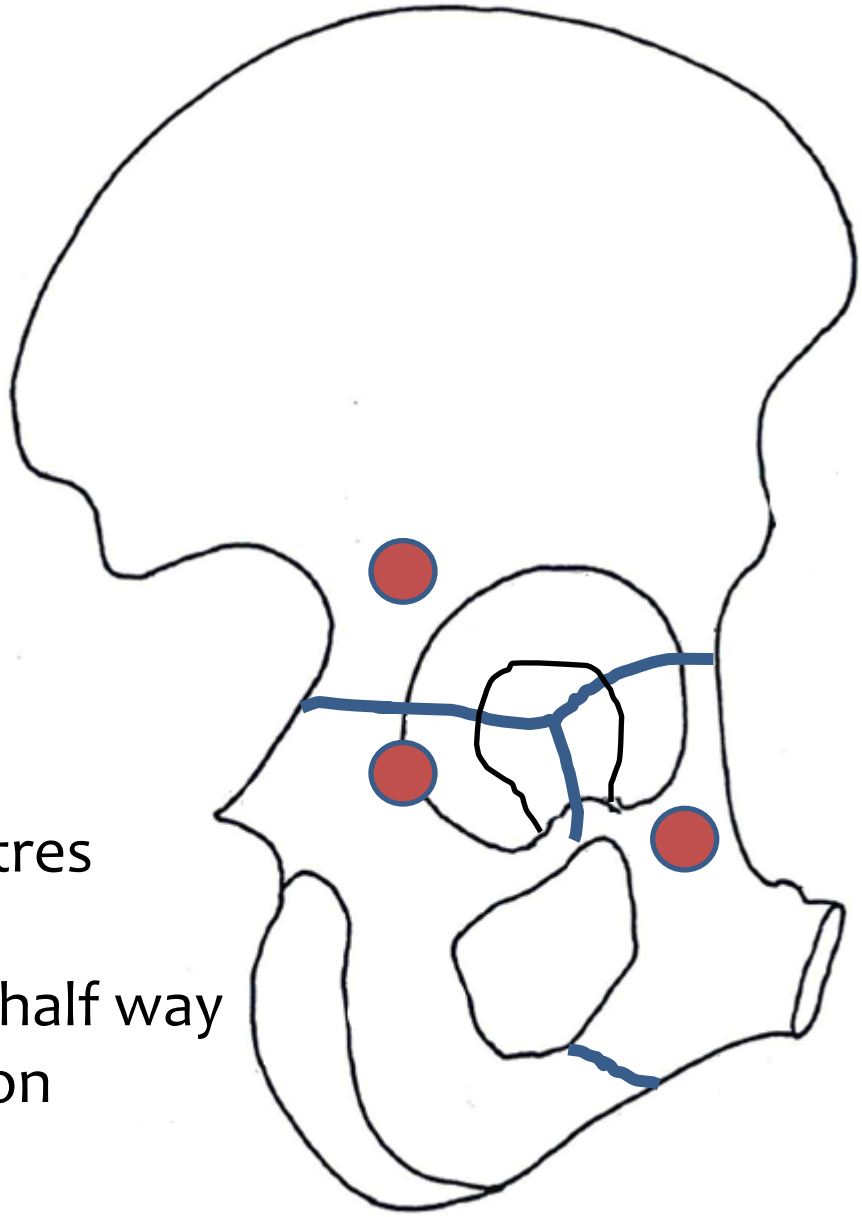
hyaline cartilage

1= lunate articular surface; 2= fibrofatty tissue called pulvinar;

3 =cotyloid fossa ; 4=labrum; 5= acetabular notch ; 6= transverse acetab. ligament

Three fused bones of Pelvis

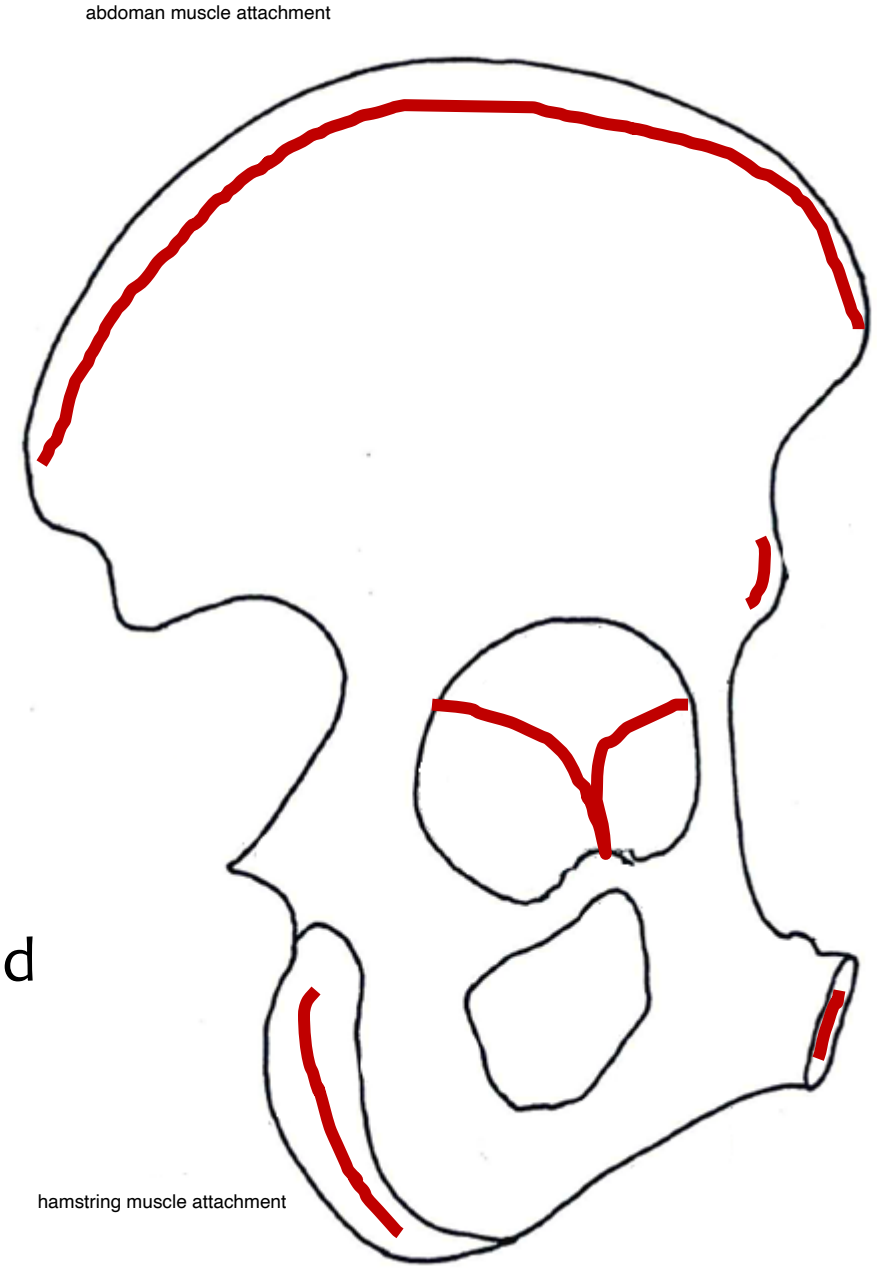


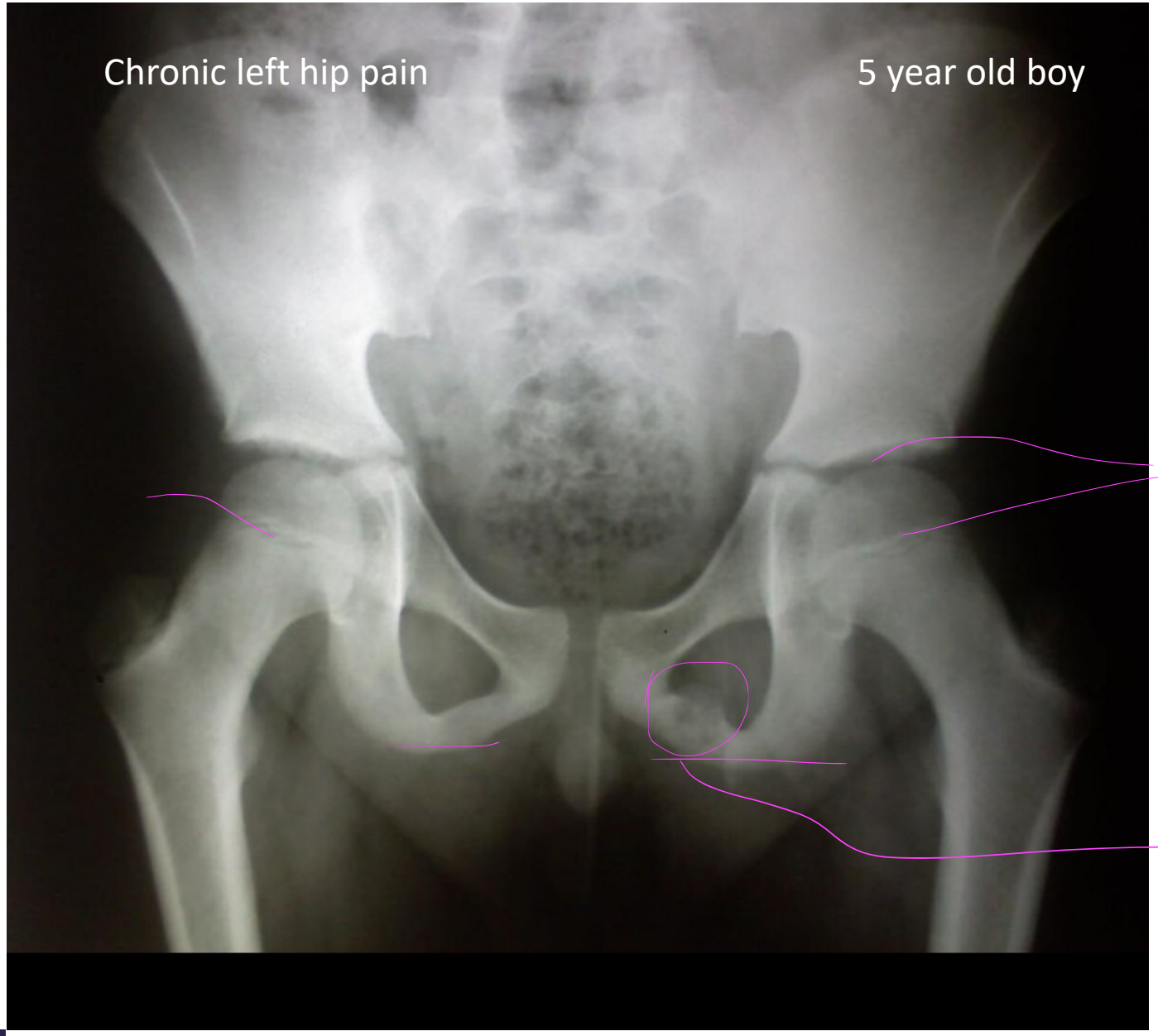


3 primary
Ossification centres

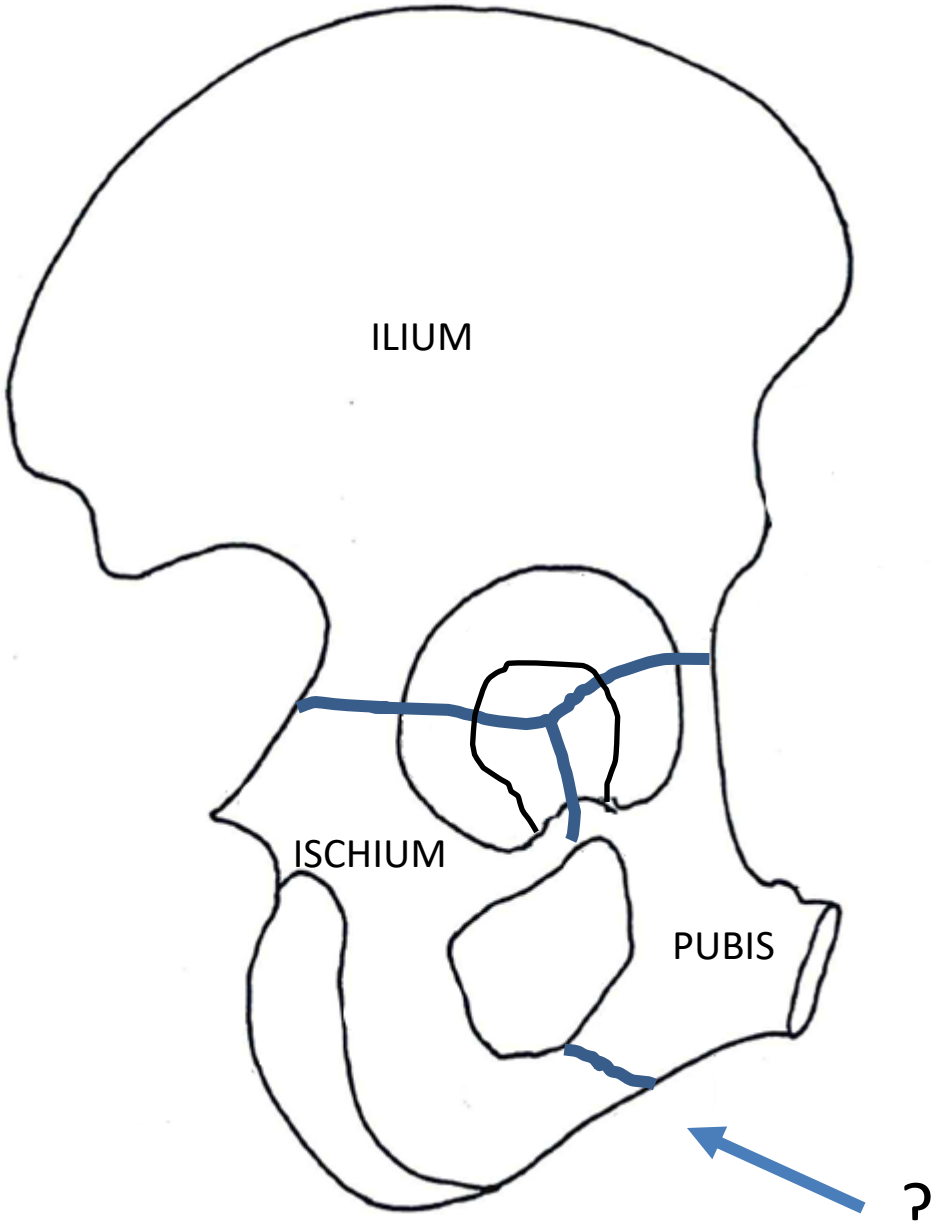
Appear approx. half way through gestation

5 secondary ossification centres in red





Radiopaedia

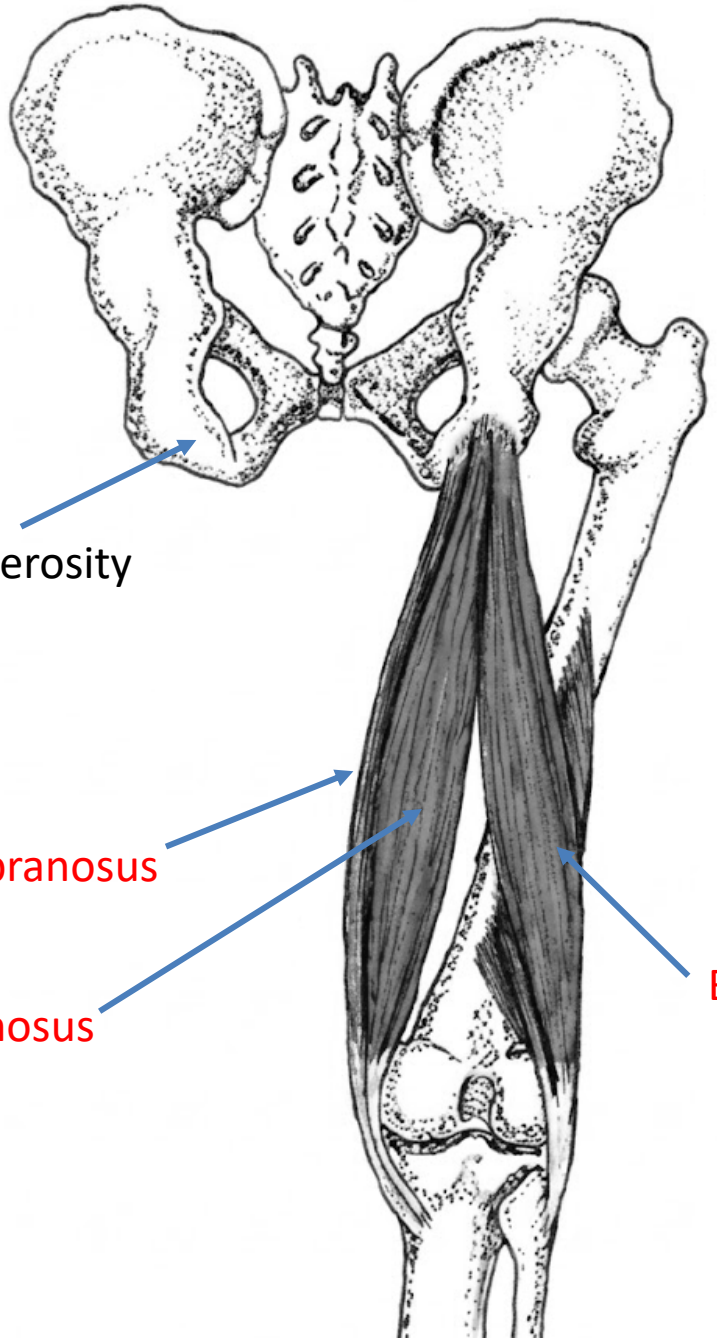

ILIUM
ISCHIUM
PUBIS
?

The **ischiopubic synchondrosis (IPS)** is a cartilaginous junction



Radiopaedia

Right Posterior view of Pelvis and Thigh



Ischial tuberosity

Semimembranosus

half muslce half membranous

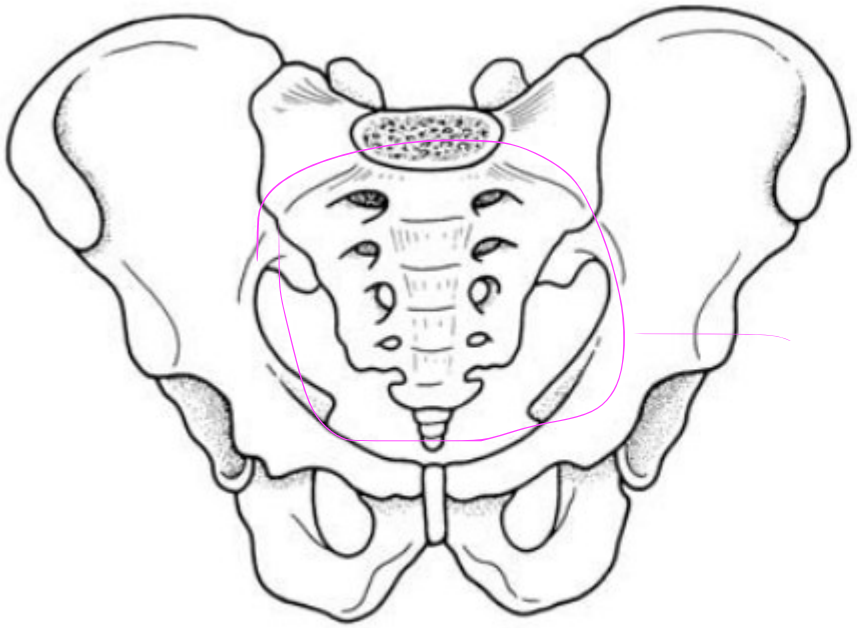
Semitendinosus

half muscle half tendon

Biceps Femoris

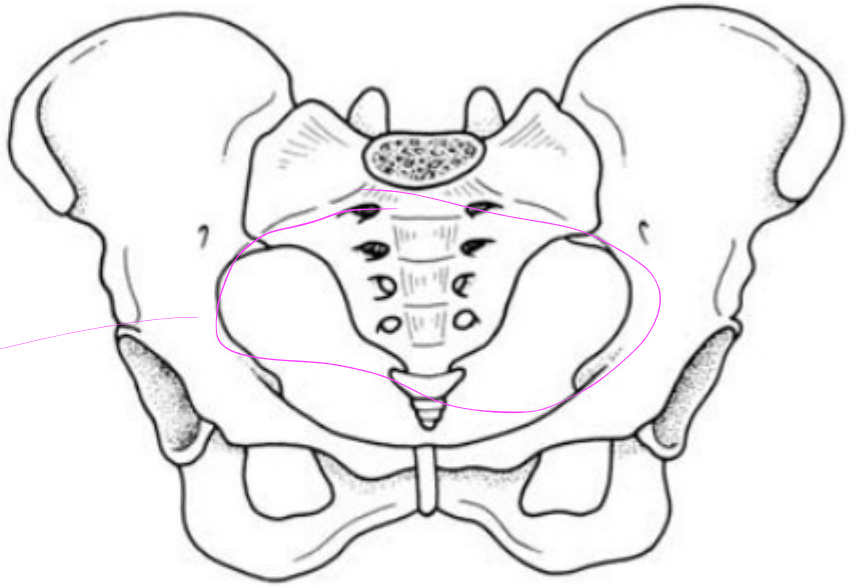
Sexual dimorphism

MALE



the inlet is broader in female

FEMALE



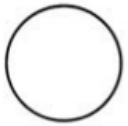
Morphology of the pelvic inlet

quite diverse shapes

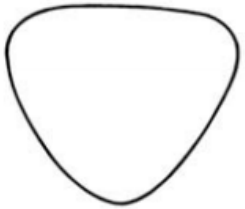


(a) Gynaecoid

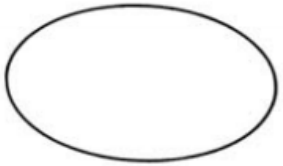
50% females have this configuration



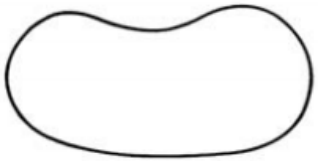
(b) Generally contracted



(c) Android



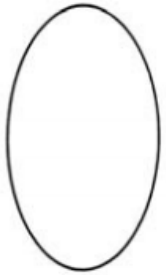
(d) Platypelloid



(e) Rachitic

Related to rickets

vit D deficiency



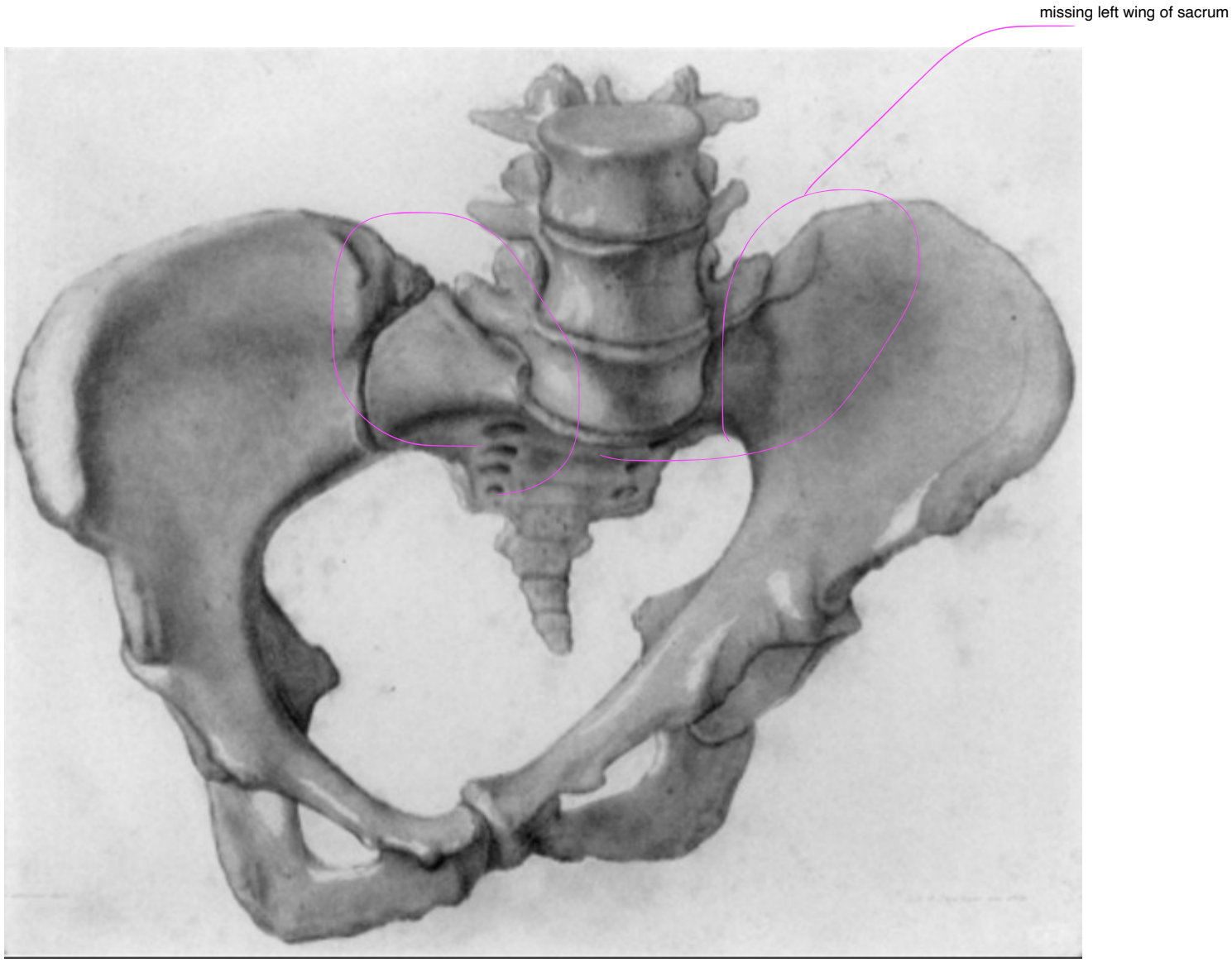
(f) Anthropoid



(g) Asymmetrical

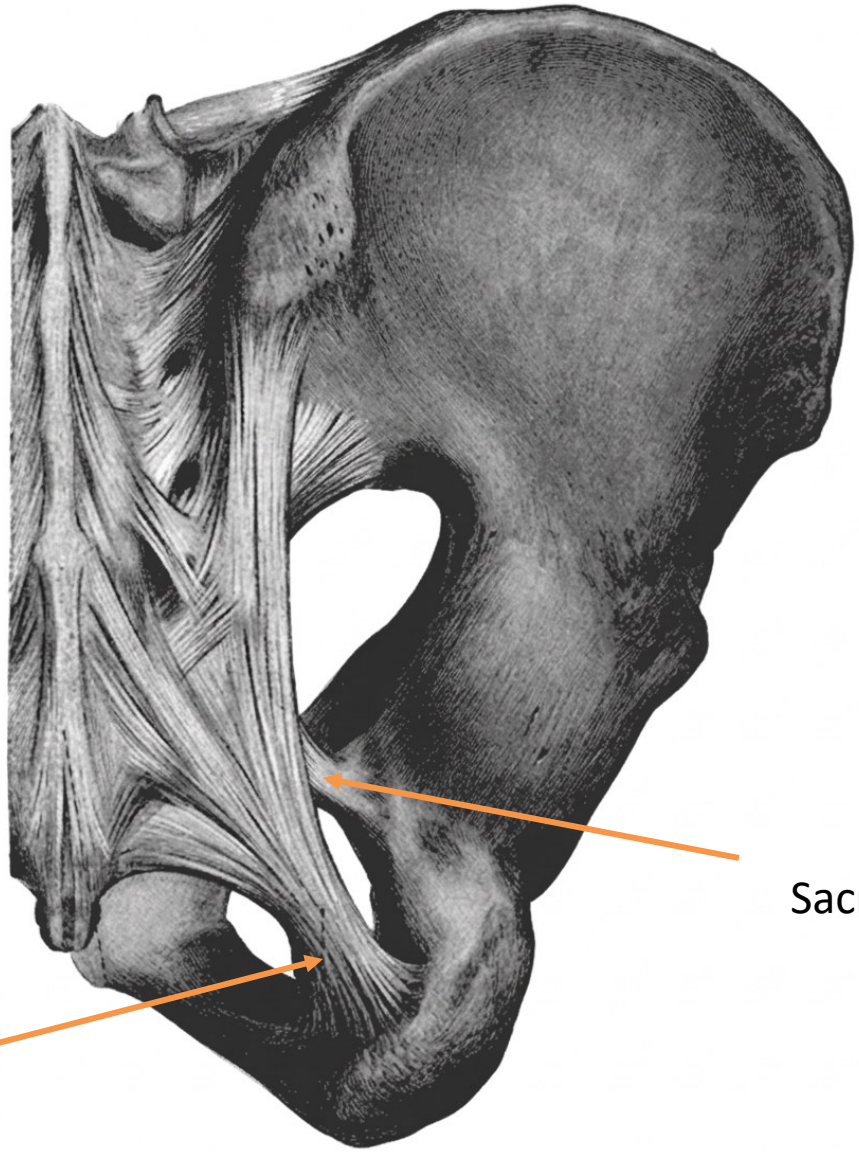
Caldwell-Moloy classification 1933

The asymmetrical Naegele Pelvis



Posterior View

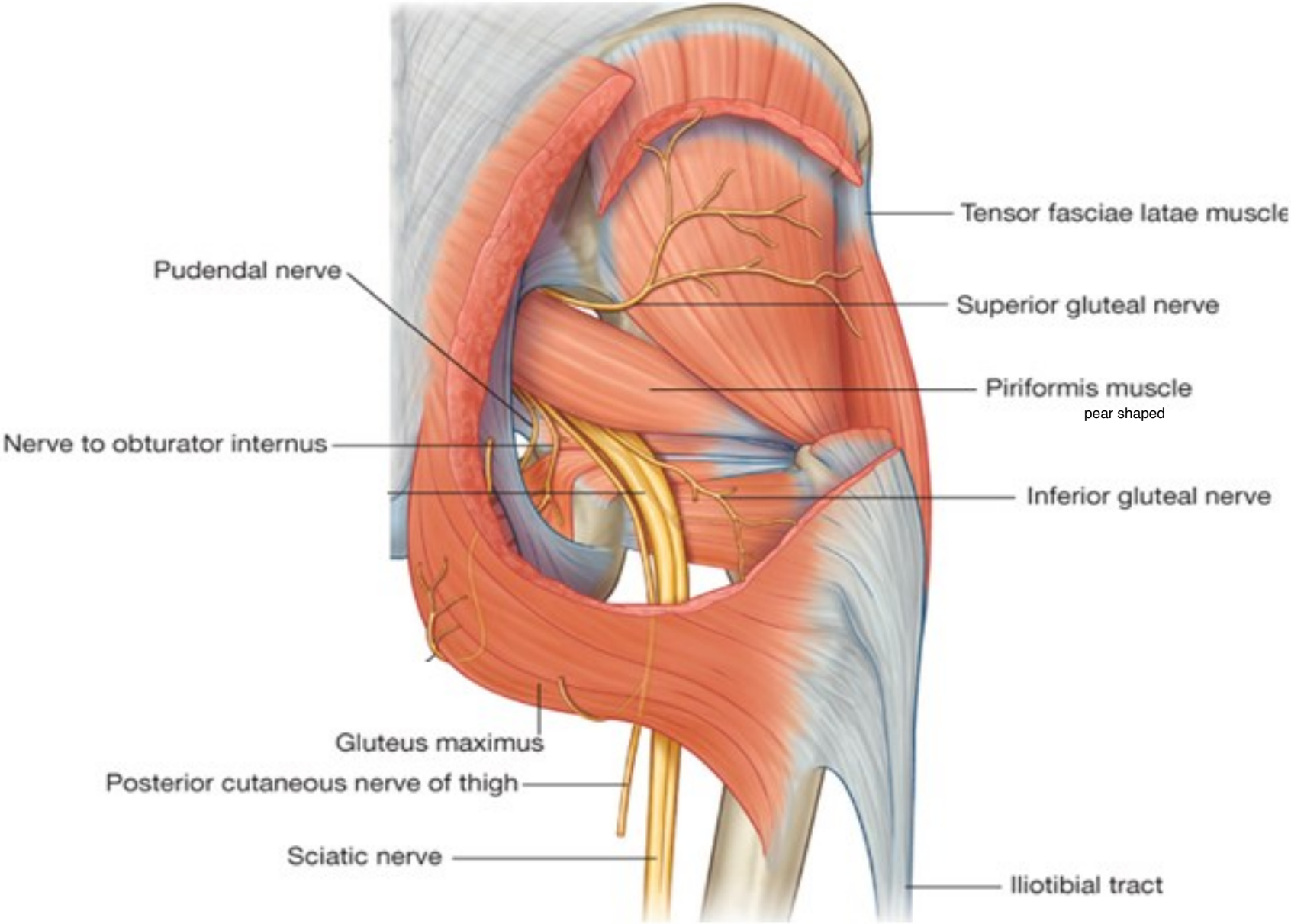




Sacrospinous ligament

Sacrotuberous ligament

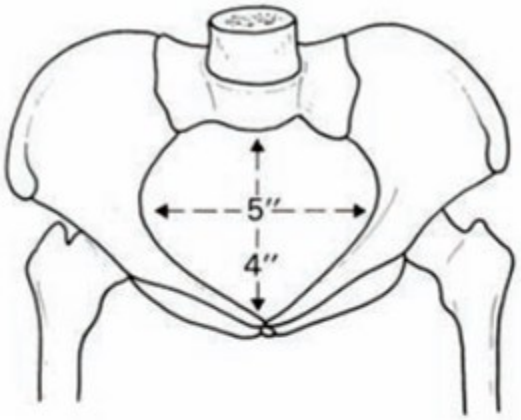
Gluteal region

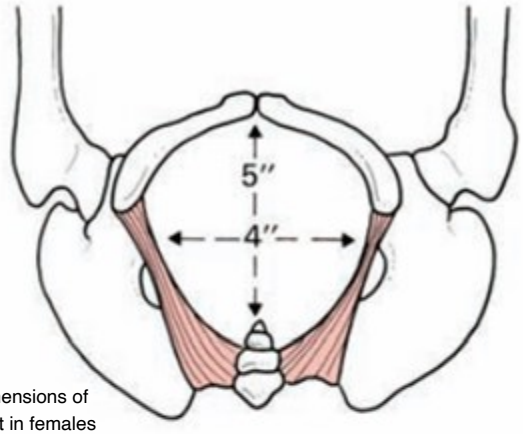


Gray's Anatomy for students

inlet

outlet

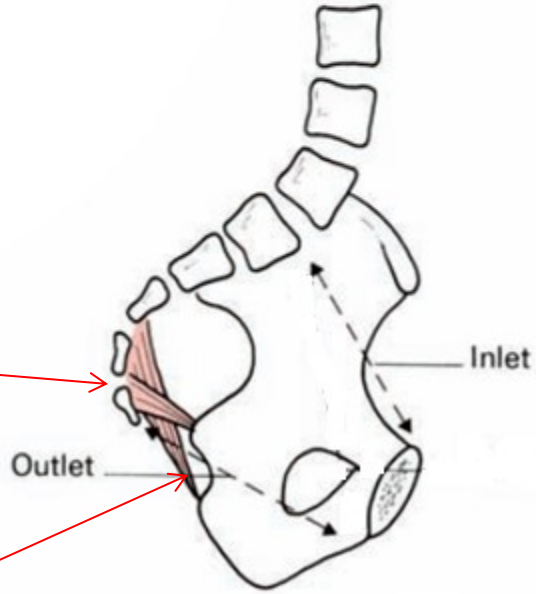




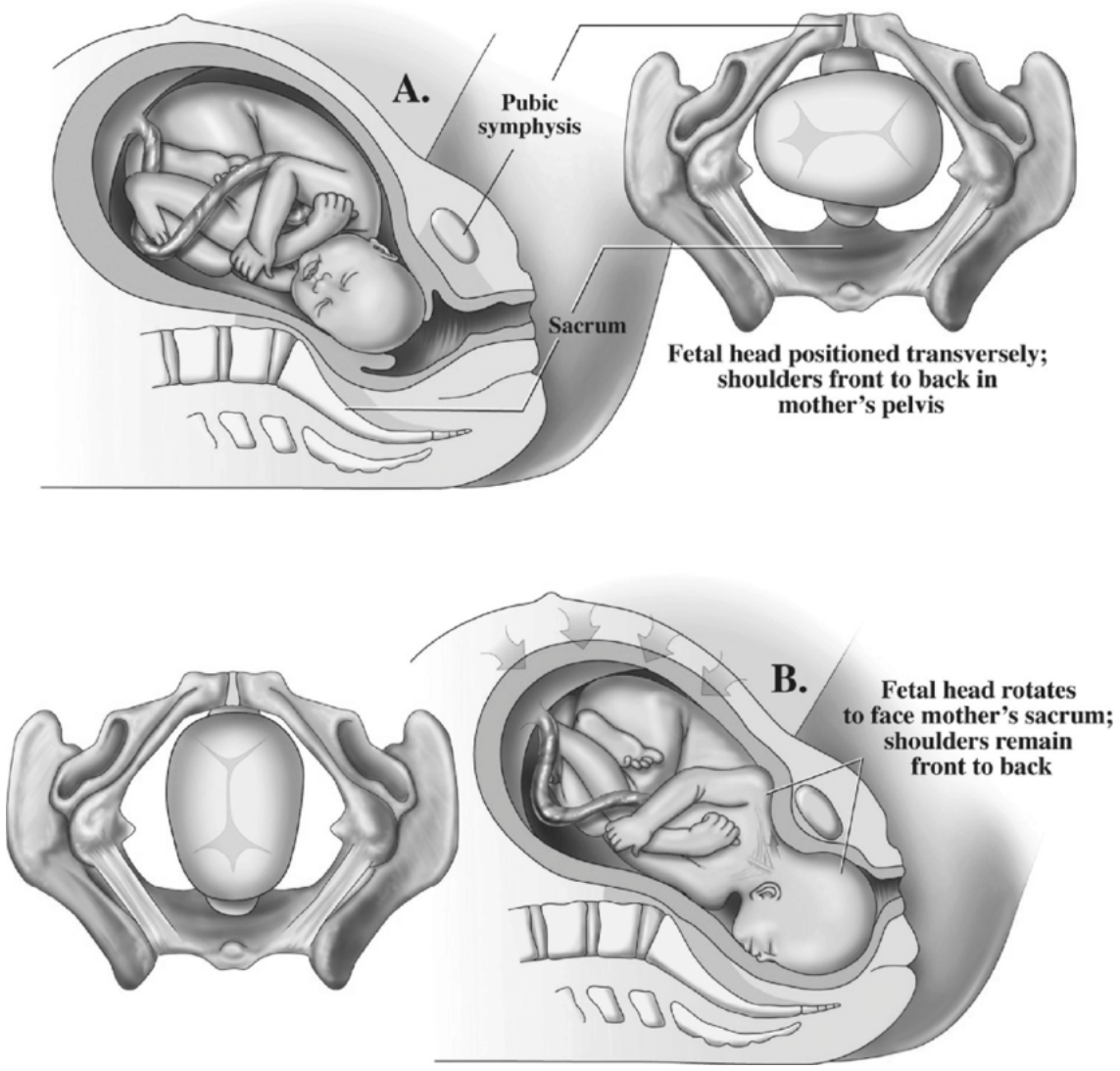
the ralive dimensions of inlet and outlet in females are inverted due to babies heads during pregnancy in the next slide

Sacrospinous ligament

Sacrotuberous ligament

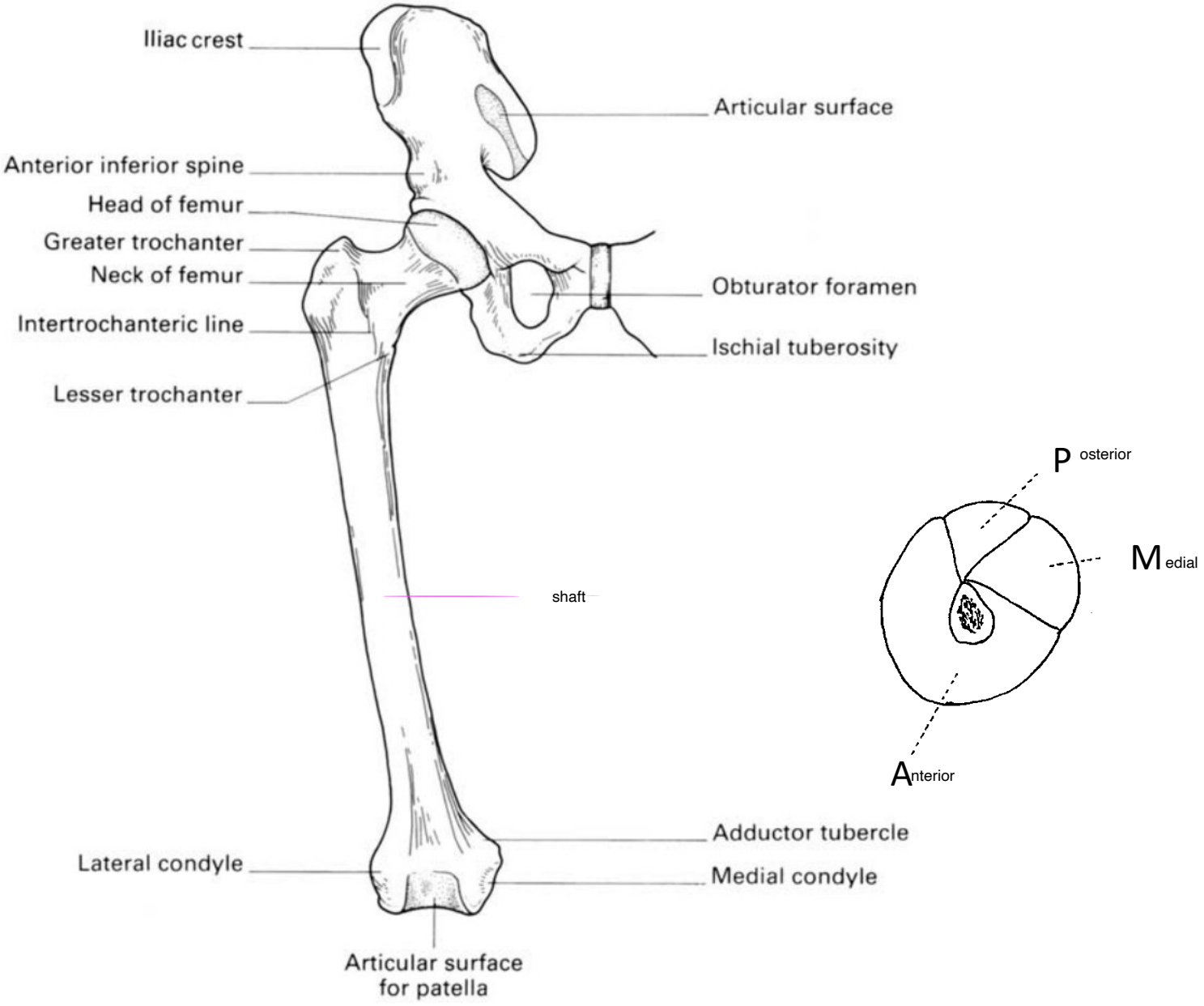


Descent of fetal head during second stage of labour

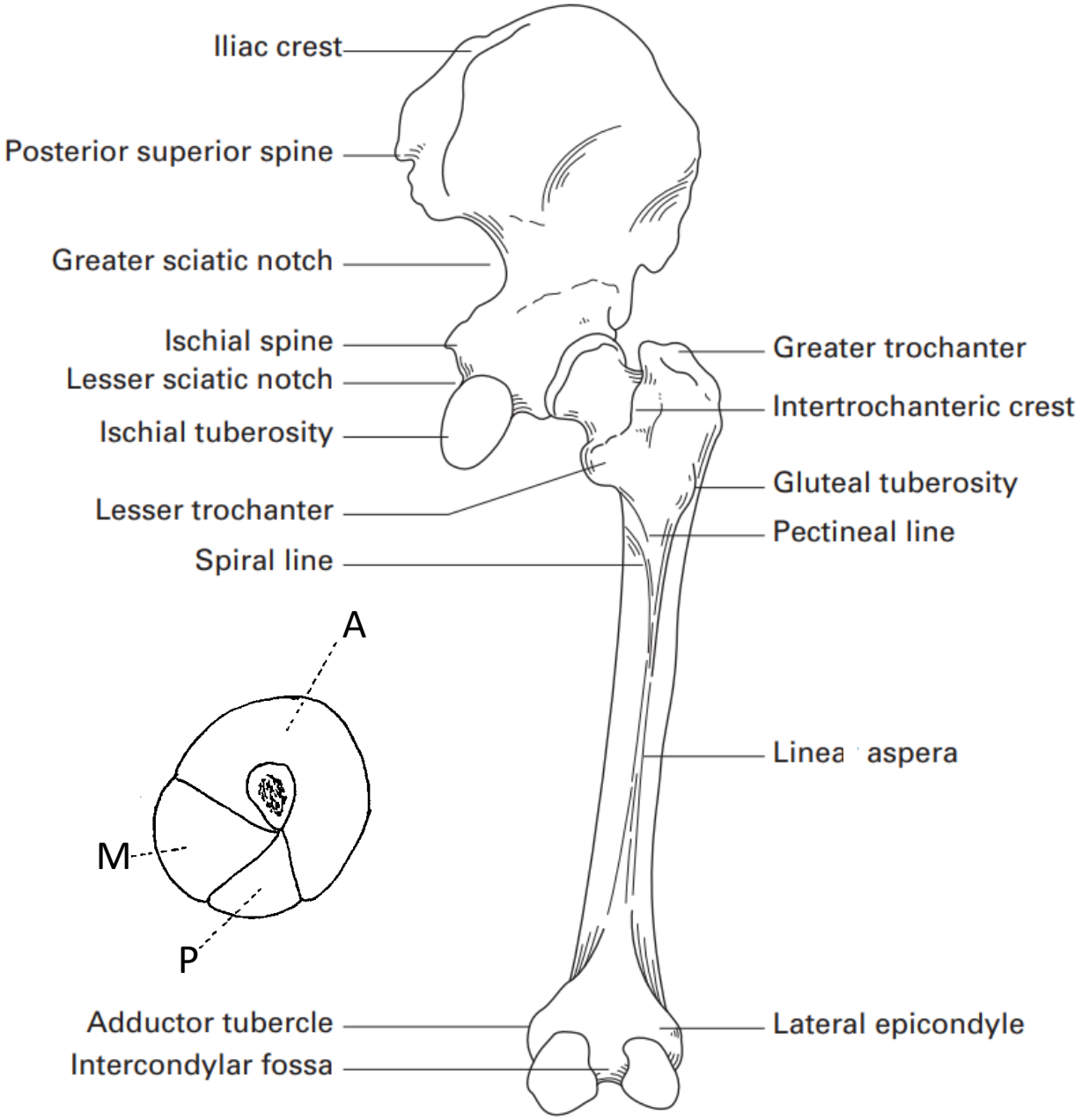


Sandmire, Herbert & Morrison, John & Racinet, Claude & Hankins, G & Pecorari, D & Gherman, R. (2008). Newborn brachial plexus injuries: The twisting and extension of the fetal head as contributing causes. Journal of obstetrics and gynaecology : the journal of the Institute of Obstetrics and Gynaecology. 28. 170-2. 10.1080/01443610801913053.

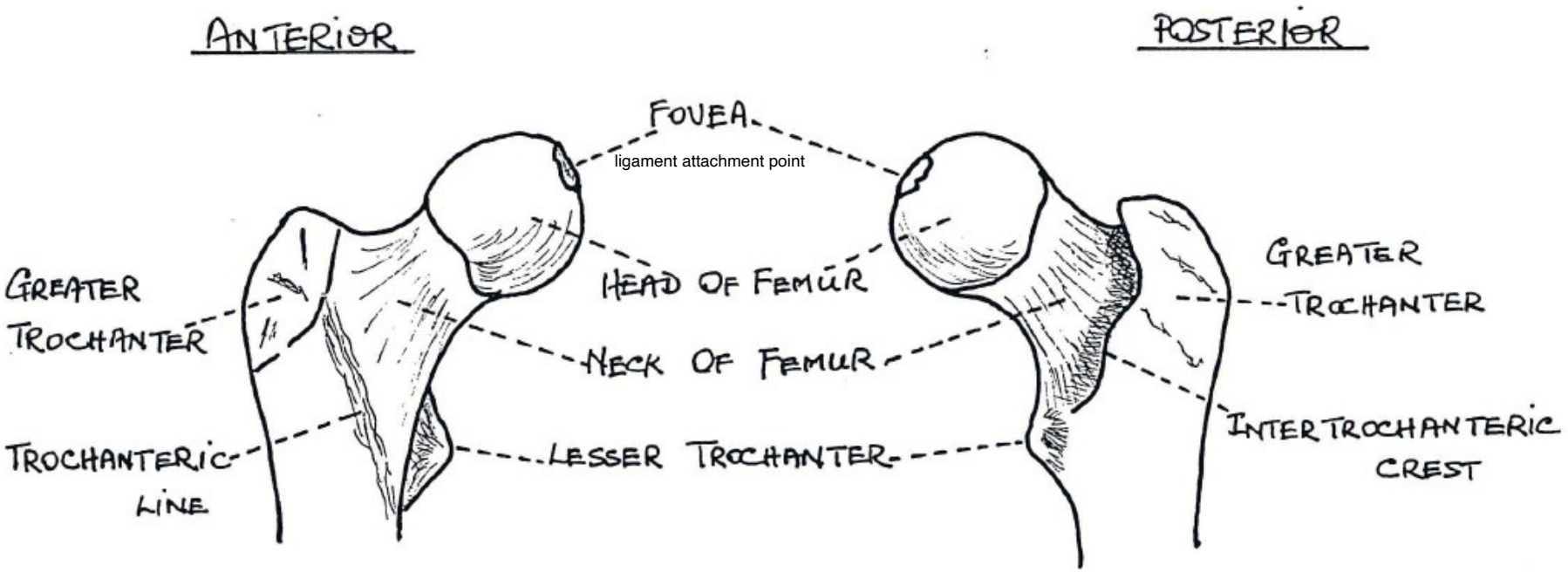
ANTERIOR VIEW



POSTERIOR VIEW

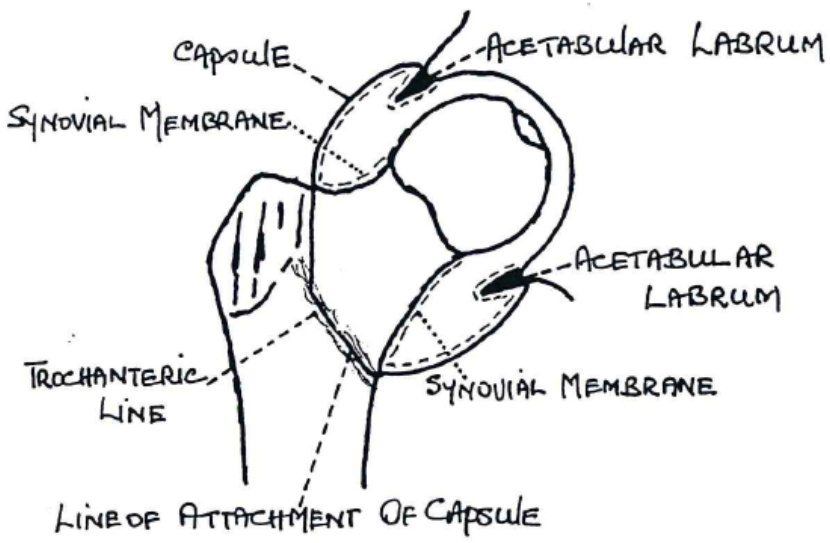


Head and neck of Femur

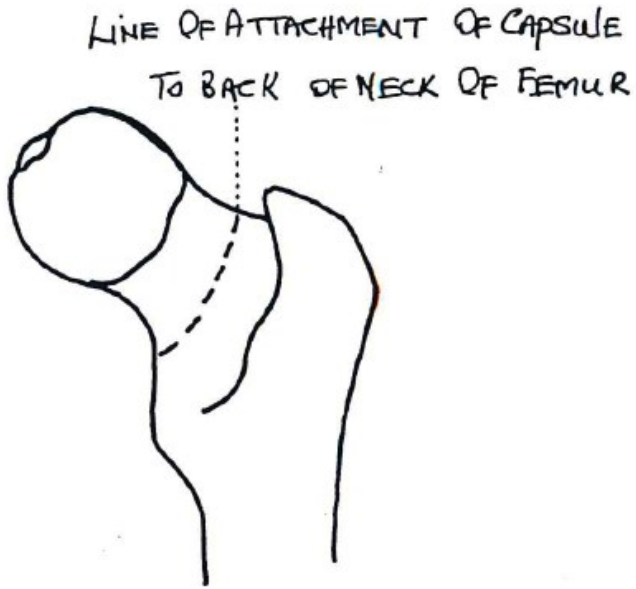


Hip Capsule

Anterior View

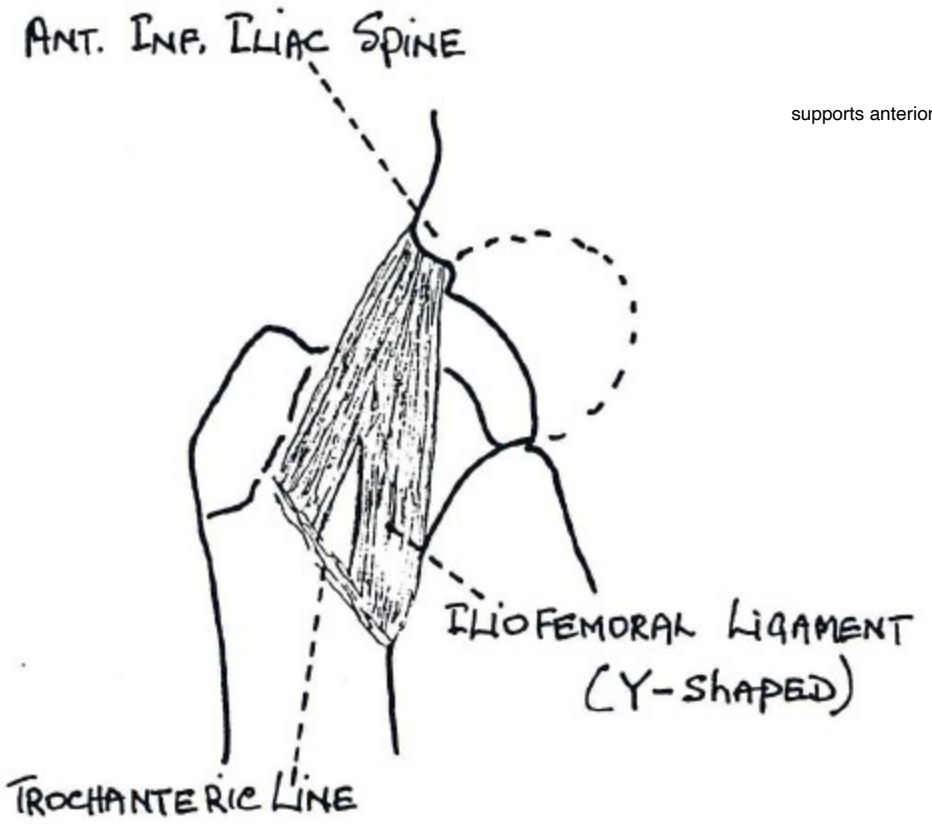


Posterior View



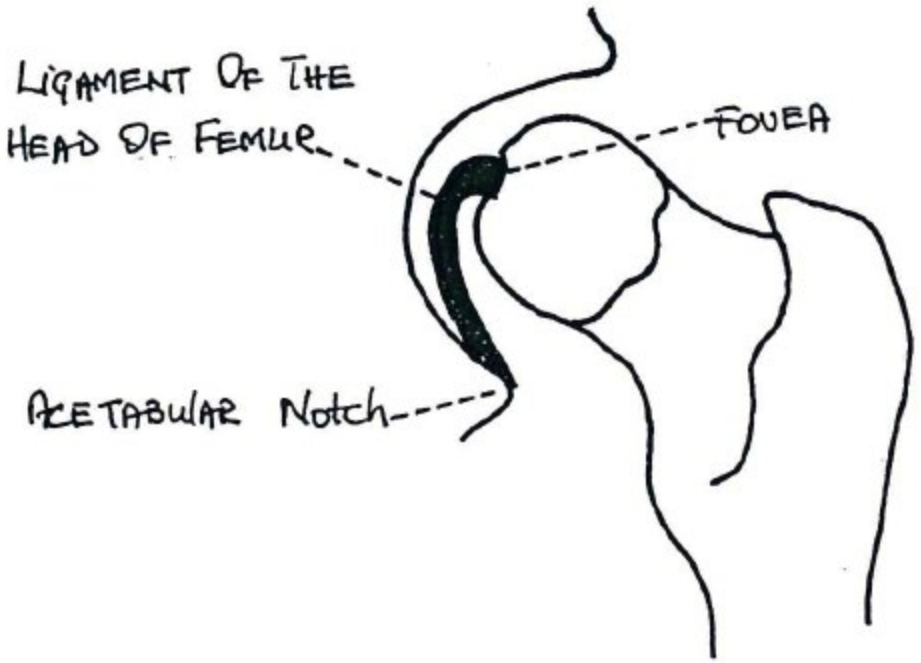
3 ligaments form the joint capsule: iliofemoral ligament
ischiofemoral ligament
pubofemoral ligament

The Iliofemoral Ligament

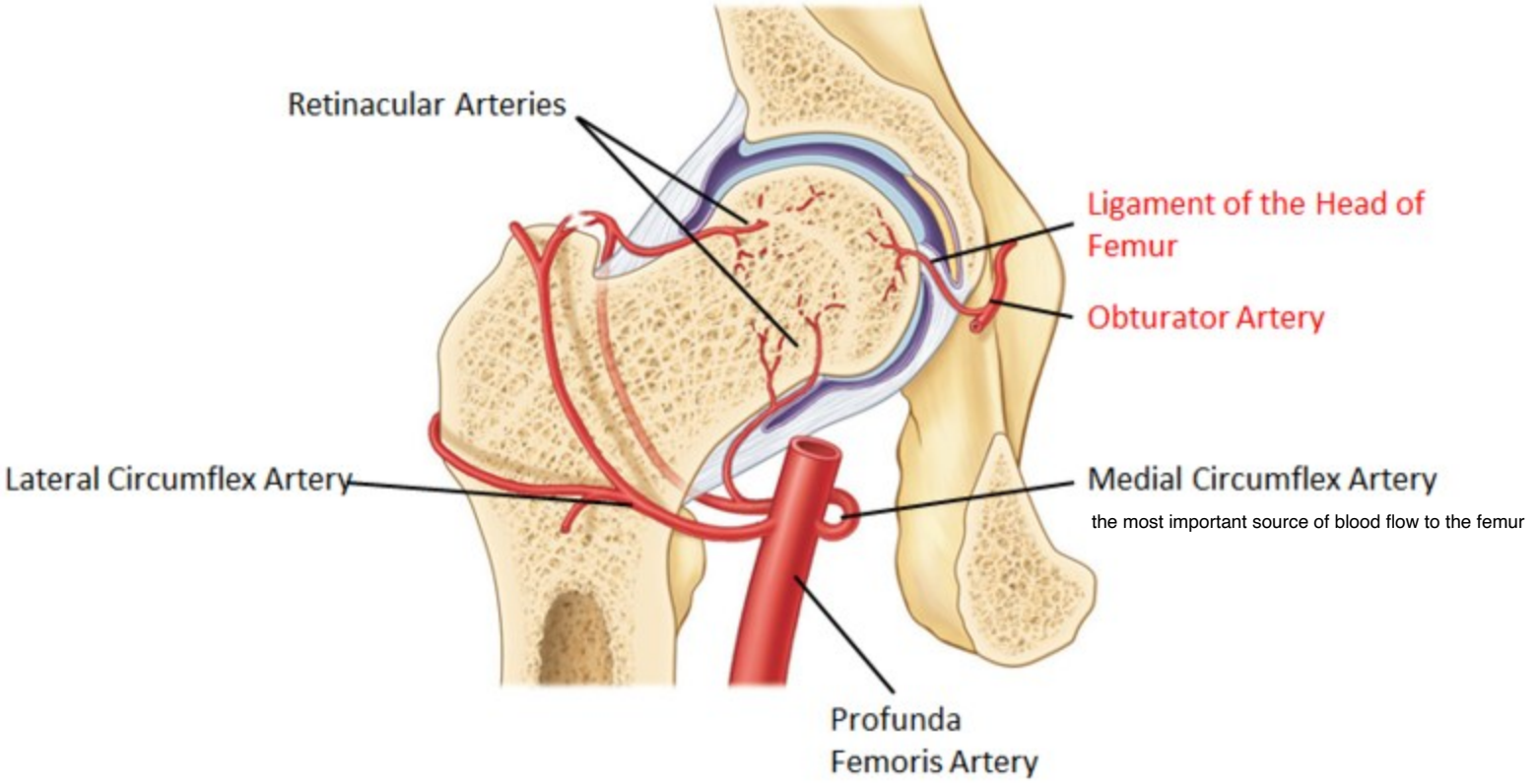


supports anteriorly to stop anterior dislocation

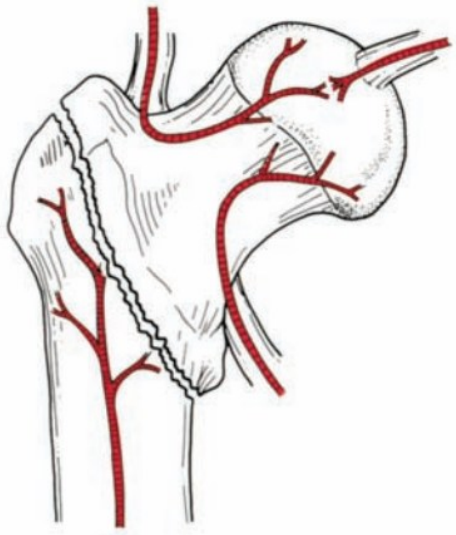
Ligamentum teres of femur



Blood supply to the head of Femur

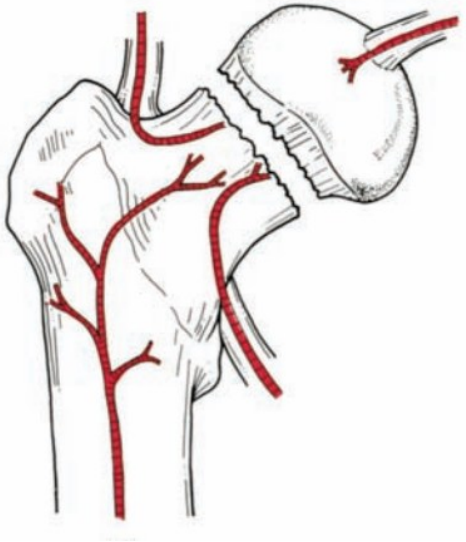


Femoral fractures



Pertrochanteric

goes through trochanters

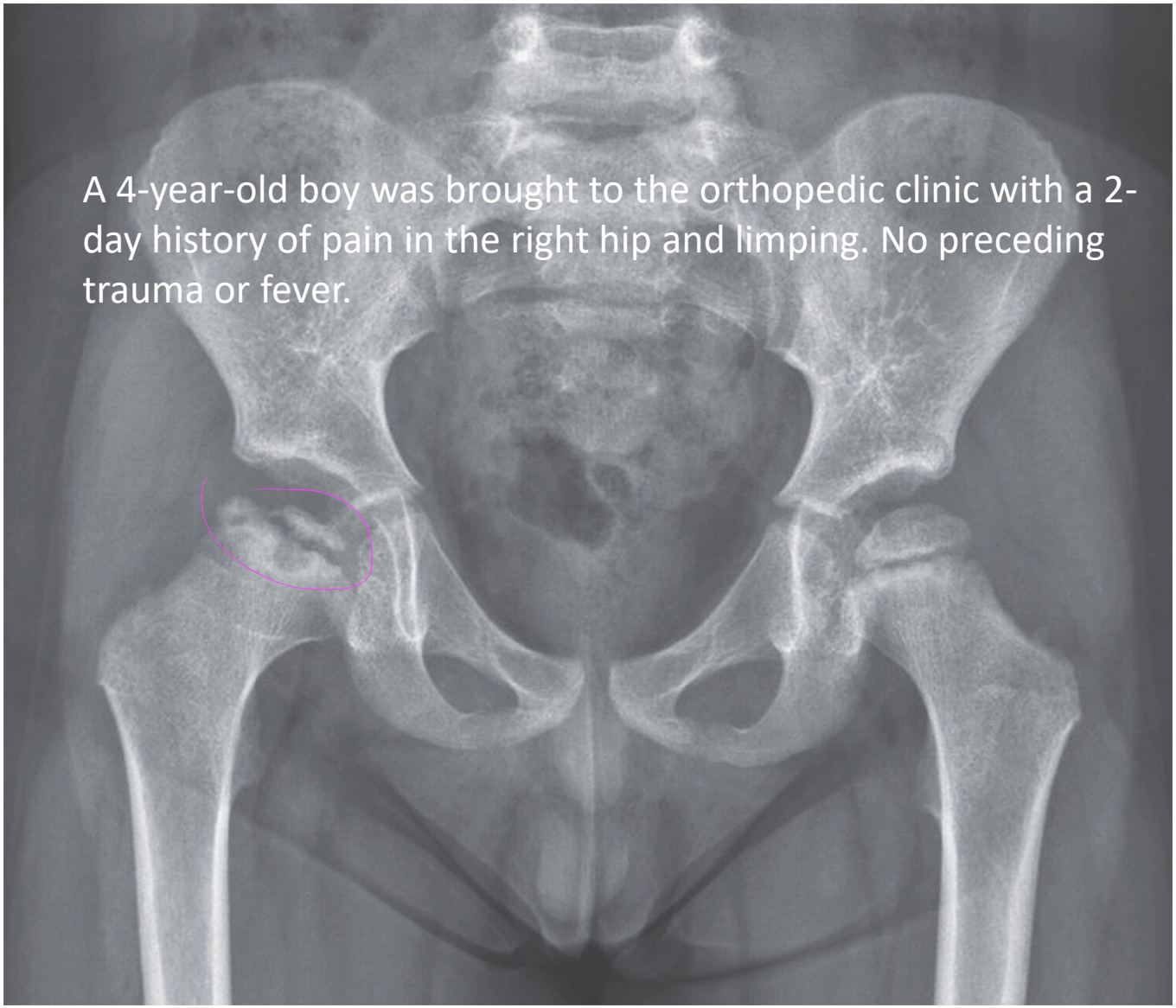


Subcapital

just under head - very paoinful and will cause necrosis

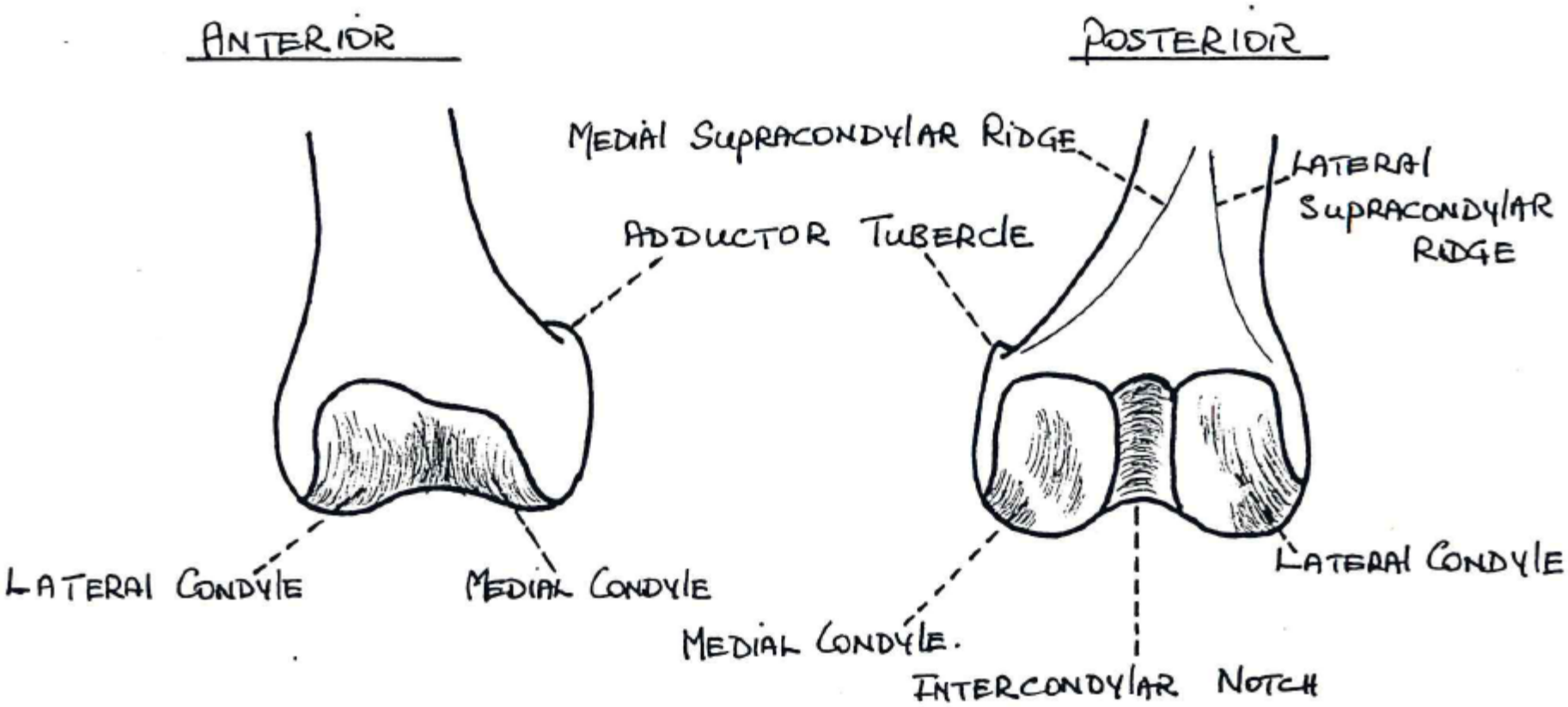
Artery of ligamentum teres (foveolar artery) is a branch of the obturator artery
The retinacular arteries are from femoral circumflex (medial and lateral), mainly medial circumflex.
(like circumflex humeral system of the upper limb)

Legg-Calve-Perthes Disease

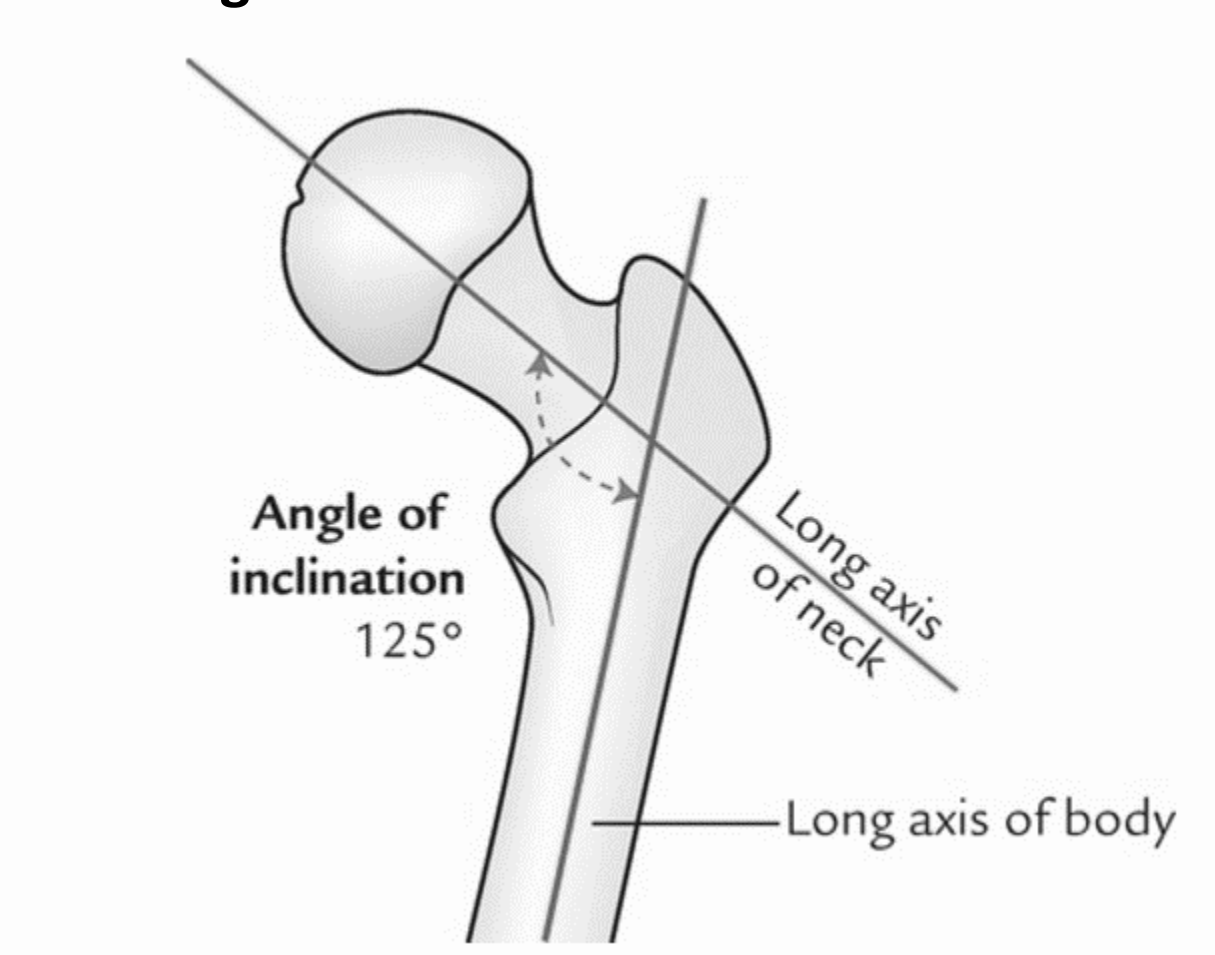


spontaneous death of head of femur

Distal end of femur



Angle of inclination of the femur



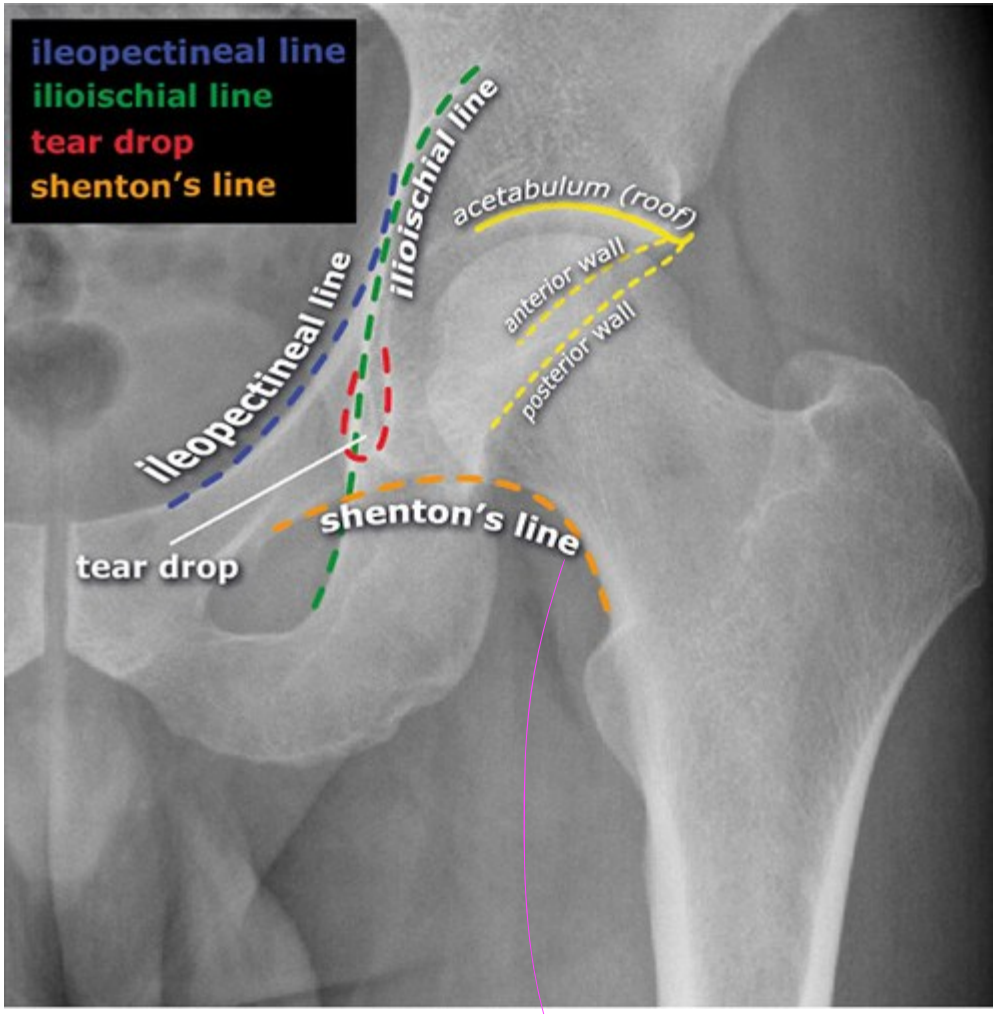
The normal neck–shaft angle is 125° in adults and 160° in children (180 ° at birth)

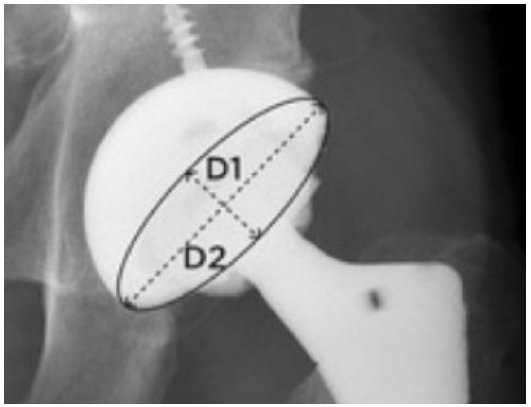
vara- push limb to midline
valga = push limb away from midline

Coxa vara	angle decrease	shaft of femur closer to midline (knock knees)
Coxa valga	angle increase	shaft of femur away from midline (bowlegged)

StartRadiology

Hip X-ray

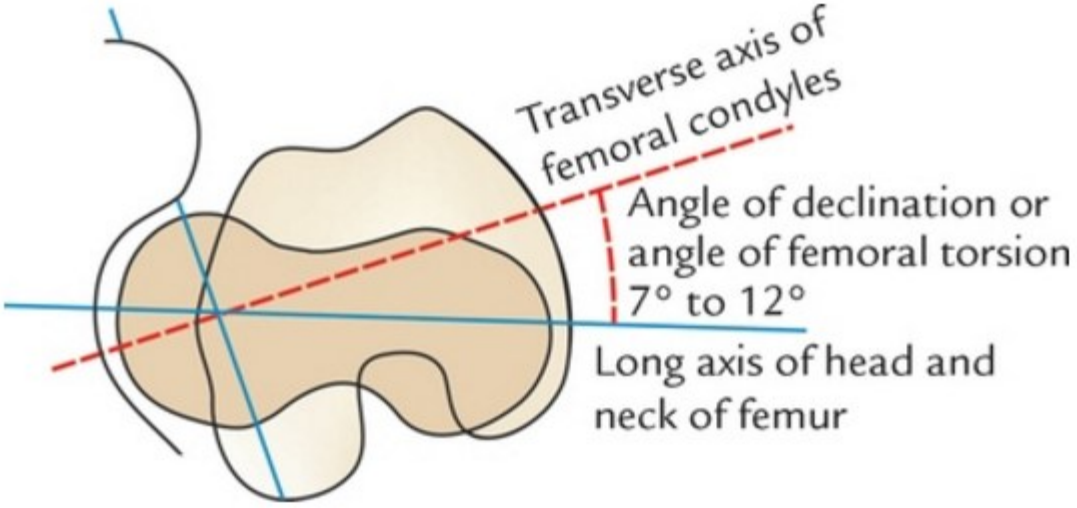




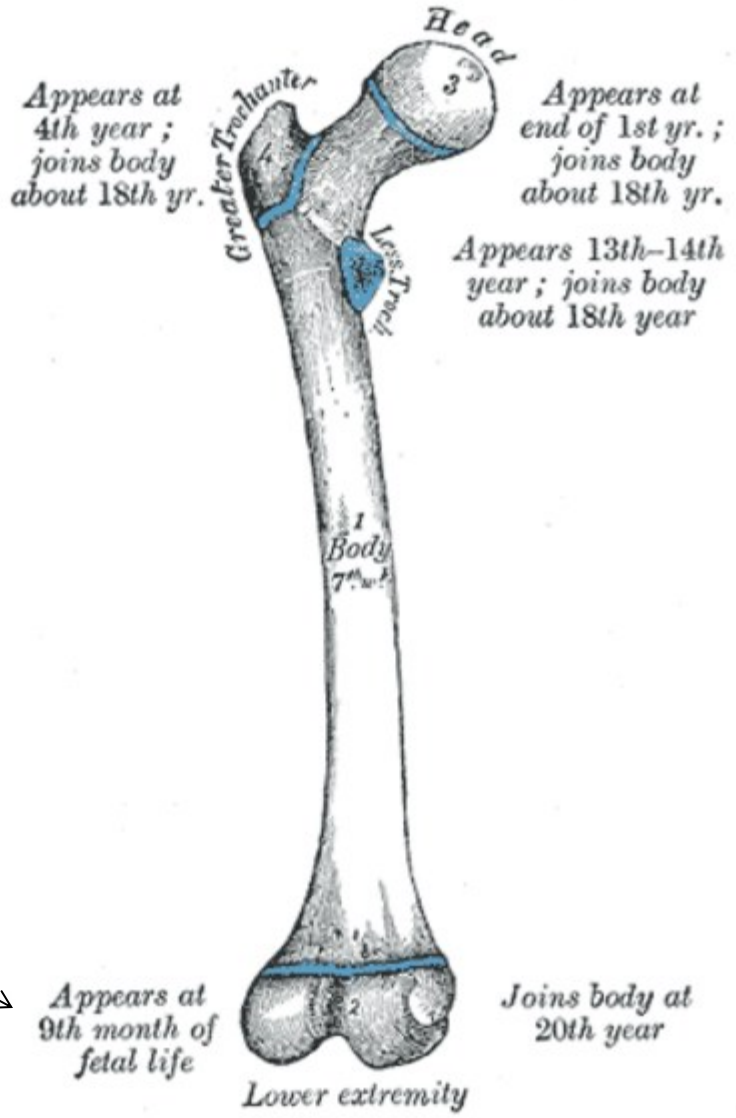
Posterior acetabular wall is more lateral because the acetabulum is anteverted by approx. 20 degrees

should be smooth, the head of the femur with the inferior edge of the superior pubic ramus

Looking down from above
we see the angle of femoral torsion

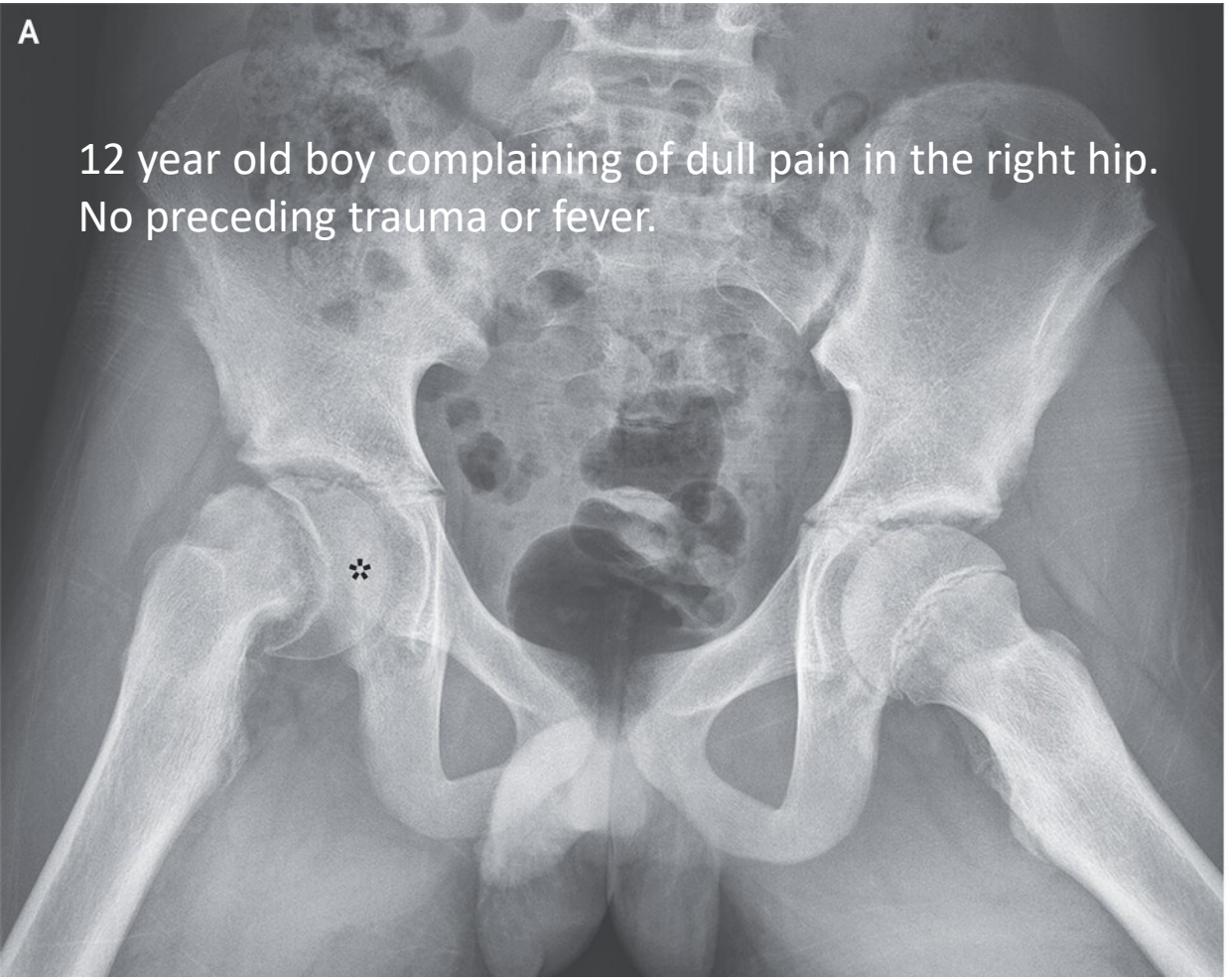


Ossification of femur



*Appears just before birth
Indicates maturity of fetus
(Medicolegal
Importance
In questions of fetal viability)

showa that the fetus is ready for birth, it has grown properly

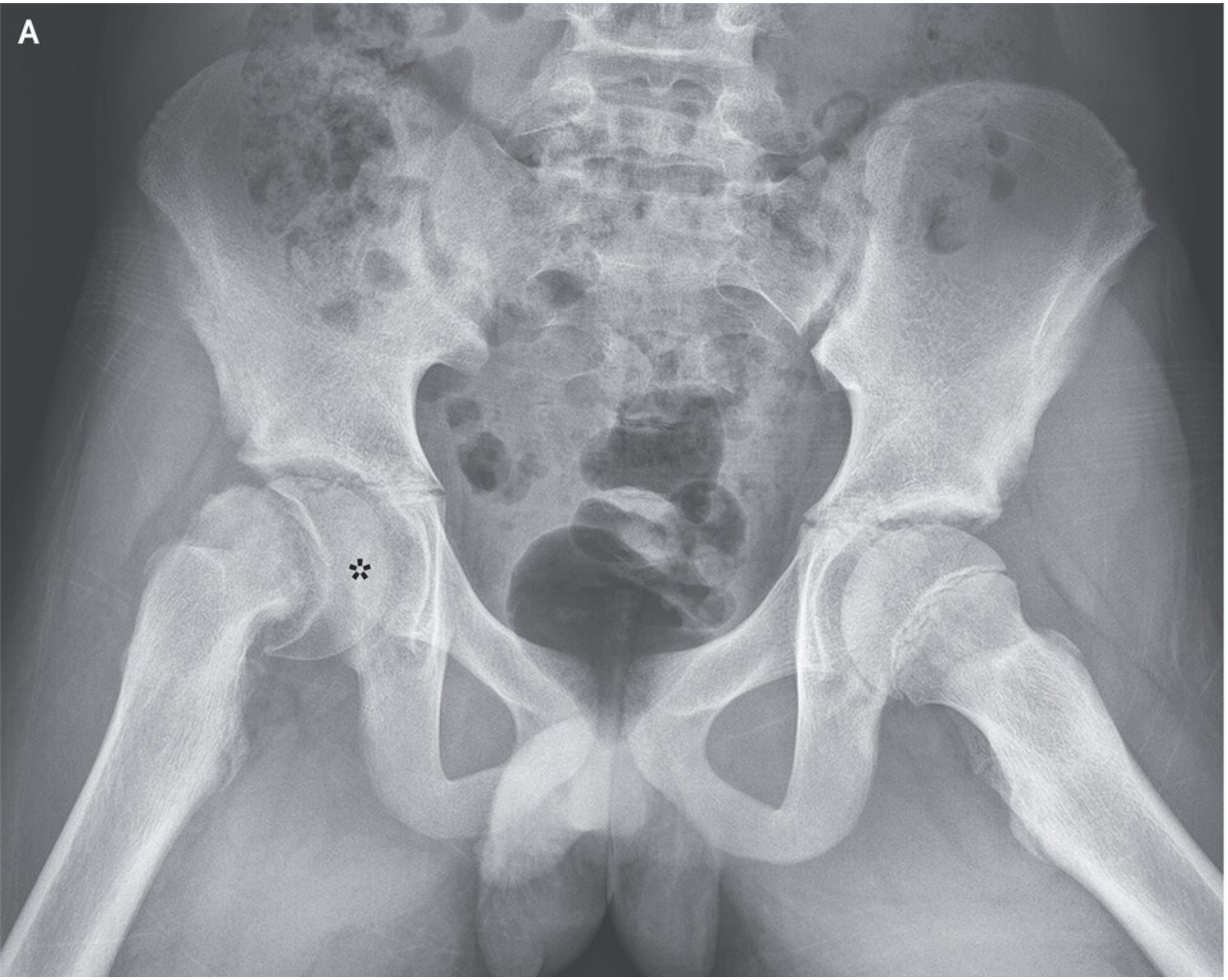


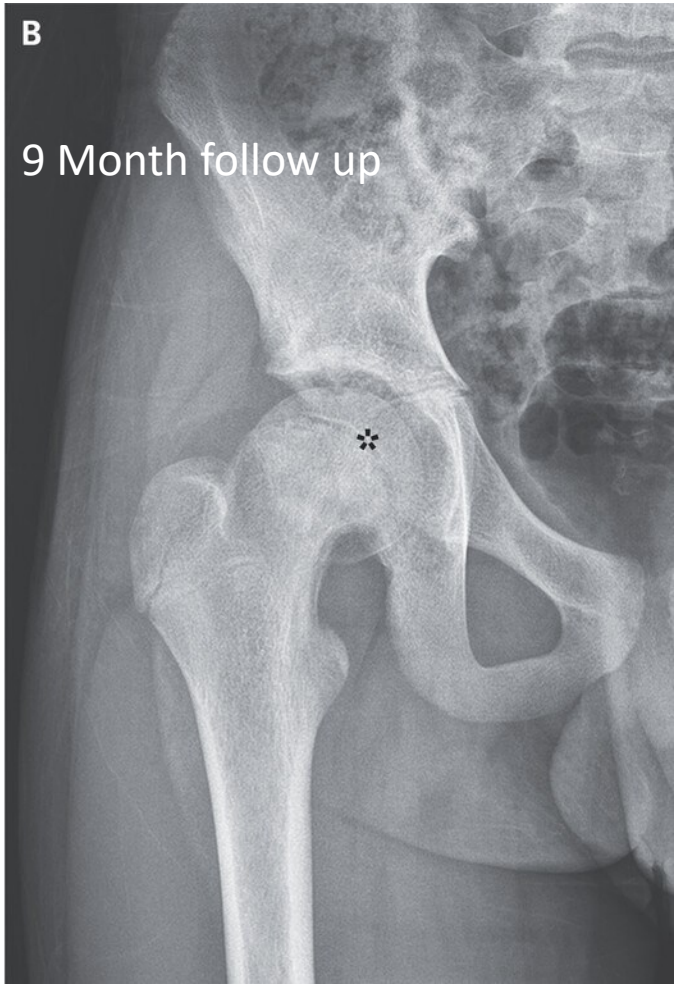


X-ray shows
a slipped ice cream cone sign *

This condition is the most common hip disorder in teenagers

Slipped capital femoral epiphysis (SCFE)

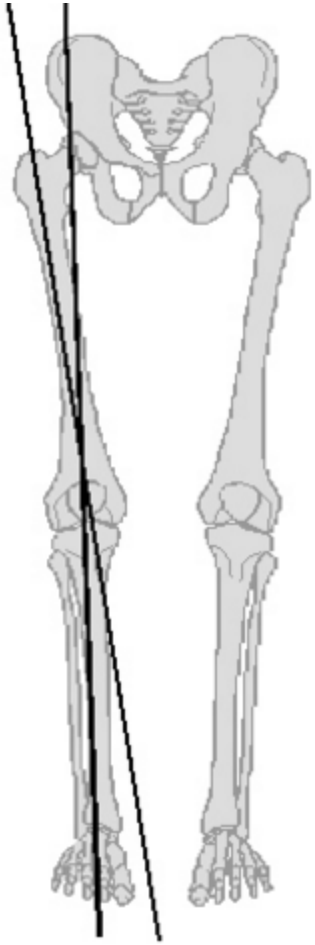




In situ fixation of the right hip was performed. At a 9-month follow-up visit, the patient's symptoms had resolved.

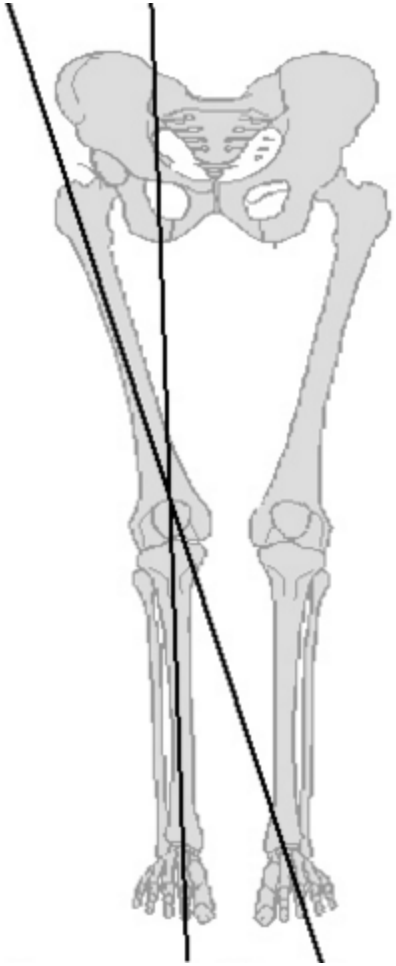
Q angle or Quadriceps angle

extensors of knee



Male

14 degrees



Female

17 degrees

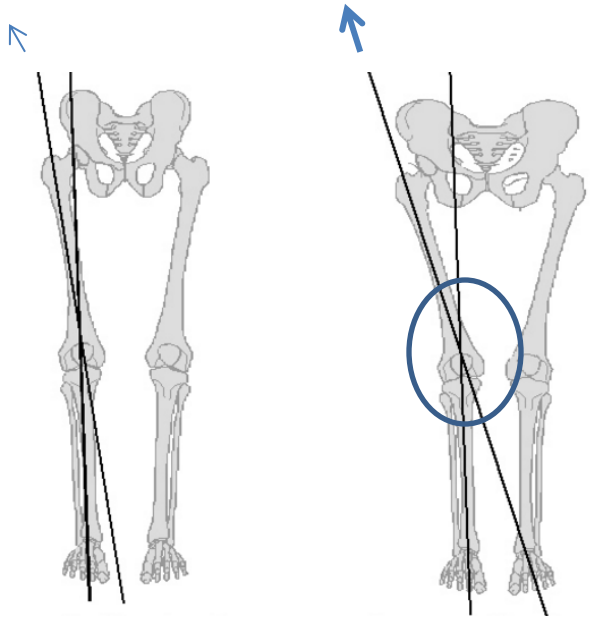
Use of Goniometer to measure Q angle



Large Q angle is a risk factor for patella subluxation



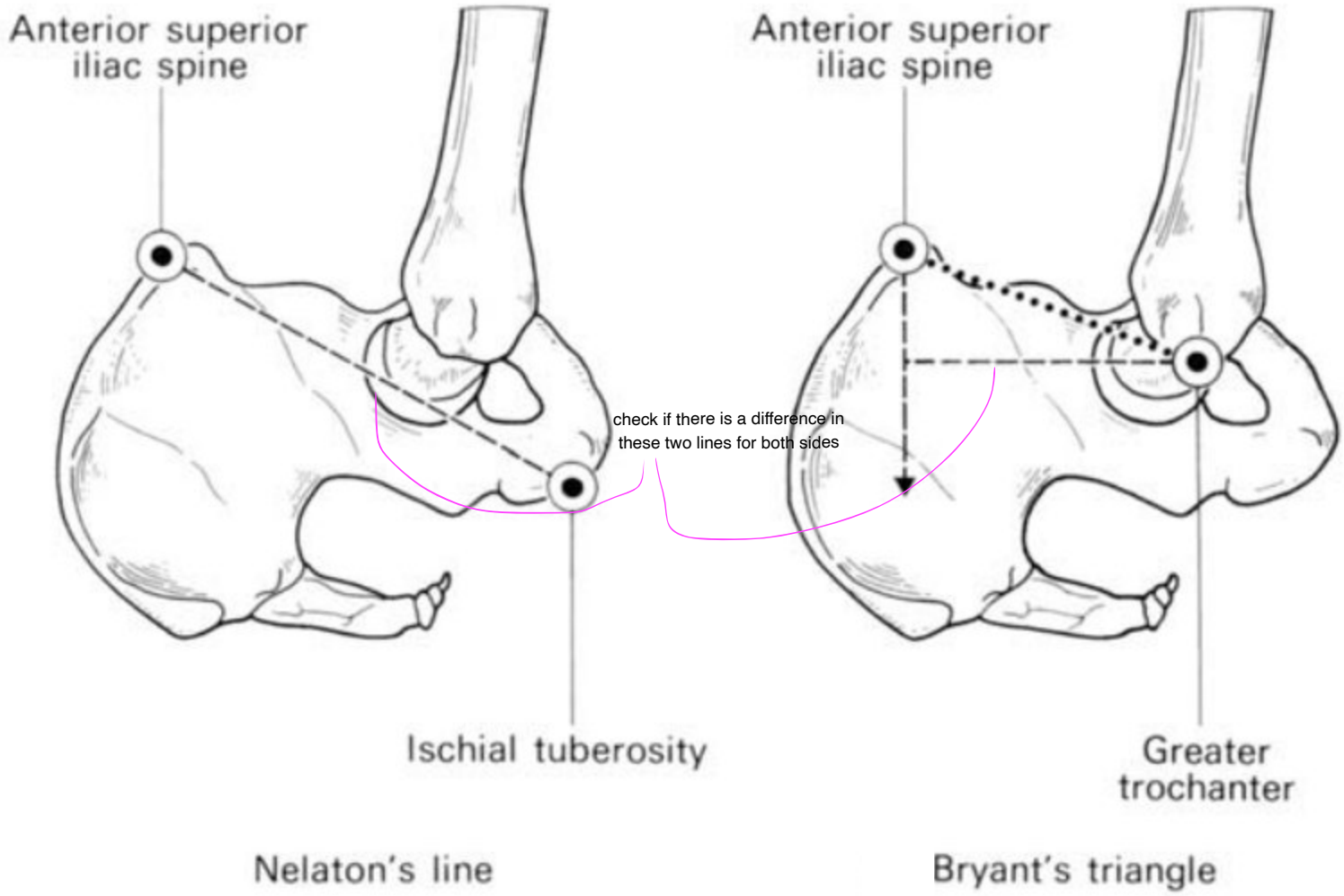
Knee AP



Knee Lat.

Lateral subluxation of right patella

Comparing left and right hips for possible femur shortening



(Or Bryant's T)

MCQ sample

In a patient who has a posterior dislocation of the hip, which of the following ligamentous structures would be torn?

A. Pubofemoral ligament
B. Iliofemoral ligament
C. Ischiofemoral ligament
D. Lacunar ligament
E. Sacrotuberous ligament